AI-Driven Multimodal Platform for Cognitive Assessment, Personalized Learning, and Vocational Readiness in Students with Down Syndrome

Project ID: 25-26J-218

Final Report

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B.Sc. (Hons) Degree in Information Technology

Specializing in Information Technology

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Final Group Report

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List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
ANN	Artificial Neural Network
NLP	Natural Language Processing
NLU	Natural Language Understanding
API	Application Programming Interface
UI/UX	User Interface/ User Experience
IoT	Internet of Things
BMI	Body Mass Index
HR	Heart Rate
SpO_2	Peripheral Capillary Oxygen Saturation
PPG	Photoplethysmography
LMS	Learning Management System
REST	Representational State Transfer
JSON	JavaScript Object Notation
NoSQL	Non-relational Database
Firebase	Google Firebase Cloud Platform
TFLite	TensorFlow Lite
CNN	Convolutional Neural Network
SVM	Support Vector Machine
RBAC	Role-Based Access Control
JWT	JSON Web Token
AES	Advanced Encryption Standard
TLS	Transport Layer Security
HTTPS	Hypertext Transfer Protocol Secure
CRUD	Create, Read, Update, Delete
KPI	Key Performance Indicator
ROI	Return on Investment
SUS	System Usability Scale
UAT	User Acceptance Testing
SDLC	Software Development Life Cycle
AR	Augmented Reality
VR	Virtual Reality
AAC	Augmentative and Alternative Communication
IEP	Individualized Education Plan
UDL	Universal Design for Learning
UNESCO	United Nations Educational, Scientific and Cultural Organization

ABSTRACT

Children with Down syndrome experience persistent challenges in communication, cognition, emotional regulation, and vocational readiness. In Sri Lanka, the lack of culturally adapted and integrated digital learning solutions limits their access to inclusive education and independent living. Existing systems are often fragmented and designed for Western contexts, failing to address the interconnected developmental needs of these learners.

This study presents the design, development, and evaluation of the Nanapiyasa Platform, an AI-driven multimodal learning system tailored for children with Down syndrome in Sri Lanka. The platform integrates four key modules: (1) a gamified cognitive and emotional assessment system using IoT-based physiological monitoring and machine learning classification; (2) an AI-powered progress tracking and parental feedback system providing real-time alerts and dashboards; (3) a personalized communication and social skills training module utilizing Natural Language Processing and a role-play chatbot; and (4) an intelligent career practice module offering behavioral profiling, vocational recommendations, and simulation-based training.

The system was developed using Flutter, Firebase, Python, and Flask, alongside a custom IoT heart rate sensor. Data were collected in collaboration with the MJF Charitable Foundation, involving approximately 600 student profiles. The classification model achieved 90.2% accuracy, while predictive analytics enabled early-warning alerts validated by experts. The chatbot demonstrated adaptive conversational capabilities, and the career module effectively matched learners with suitable roles.

User Acceptance Testing across educators, caregivers, and administrators yielded a System Usability Scale score of 80.2, indicating high usability. This research contributes a scalable, culturally responsive platform that supports inclusive education by addressing cognitive, social, physiological, and vocational development within a unified system.

Keywords

Down syndrome, inclusive education, machine learning, multimodal learning, Sri Lanka

1. INTRODUCTION

Inclusive education is enshrined as a fundamental human right under the United Nations Convention on the Rights of Persons with Disabilities (UNCRPD) and the UNESCO Salamanca Statement, mandating that all learners, irrespective of cognitive or physical ability, receive equitable, meaningful, and high-quality educational opportunities [1]. Among learner populations with complex special educational needs, children with Down syndrome present a particularly multidimensional profile of challenges, encompassing deficits in expressive language, working memory, processing speed, social communication, attention regulation, emotional control, and adaptive behaviour [2]. Down syndrome caused by the presence of a full or partial third copy of chromosome 21, a condition termed trisomy 21 is one of the most prevalent chromosomal causes of intellectual disability worldwide, affecting approximately one in every 700 to 1,000 live births globally [3]. If inadequately addressed during formative and transitional educational years, the interacting challenges associated with Down syndrome significantly constrain long-term prospects for independent living, social participation, and vocational engagement [4].

In the Sri Lankan context, the Disability Rights Act No. 9 of 1996 and the National Policy on Disability formally affirm the rights of individuals with special needs to education and equitable employment opportunities [5]. However, the practical implementation of inclusive education in Sri Lanka remains severely constrained. The country lacks specialized digital tools adapted to the cultural, linguistic, and socioeconomic realities of Sri Lankan learners with Down syndrome; there is an acute shortage of educators trained in assistive technologies; and scalable, culturally responsive learning systems are largely absent from the national educational ecosystem [6]. The overwhelming majority of available digital platforms have been developed for heterogeneous disability populations or within high-income Western educational contexts, and do not account for the linguistic, cultural, or infrastructural variables specific to Sri Lanka [7].

Recent advances in Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and Internet of Things (IoT) technologies present compelling and largely unexplored opportunities to bridge these structural gaps. Adaptive learning systems powered by AI are capable of analysing nuanced learner behaviour and dynamically adjusting instructional content to the individual's learning trajectory [8]. Predictive analytics frameworks enable the early detection of learning difficulties, attention decline, and disengagement, thereby supporting timely and evidence-informed caregiver and educator intervention [9]. NLP-driven conversational agents provide interactive, safe, and repeatable environments for the structured practice of communication and social interaction skills [8]. Gamification strategies encompassing reward mechanisms, levelled progression, and interactive challenges have been empirically demonstrated to enhance sustained engagement and intrinsic motivation in learners with intellectual disabilities [10]. Simulation-based

vocational training environments further enable learners to develop practical, job-relevant skills in controlled, consequence-free settings that bridge the gap between structured education and real-world employment [4].

Despite the growing body of evidence supporting each of these technological approaches individually, a critical examination of the existing landscape reveals that no integrated platform has been developed to simultaneously address cognitive and emotional assessment, predictive progress monitoring, communication and social skills development, and vocational readiness training for learners with Down syndrome within a single, coherent, and culturally appropriate framework particularly not within the Sri Lankan context. The present research addresses this overarching gap through the design, development, and empirical evaluation of the Nanapiyasa Platform: a unified, AI-driven, multimodal learning system integrating four interoperable modules, each targeting a distinct but interconnected dimension of holistic learner development. The following sections present the background literature, articulate the identified research gaps, define the research problem, and establish the specific objectives guiding this study.

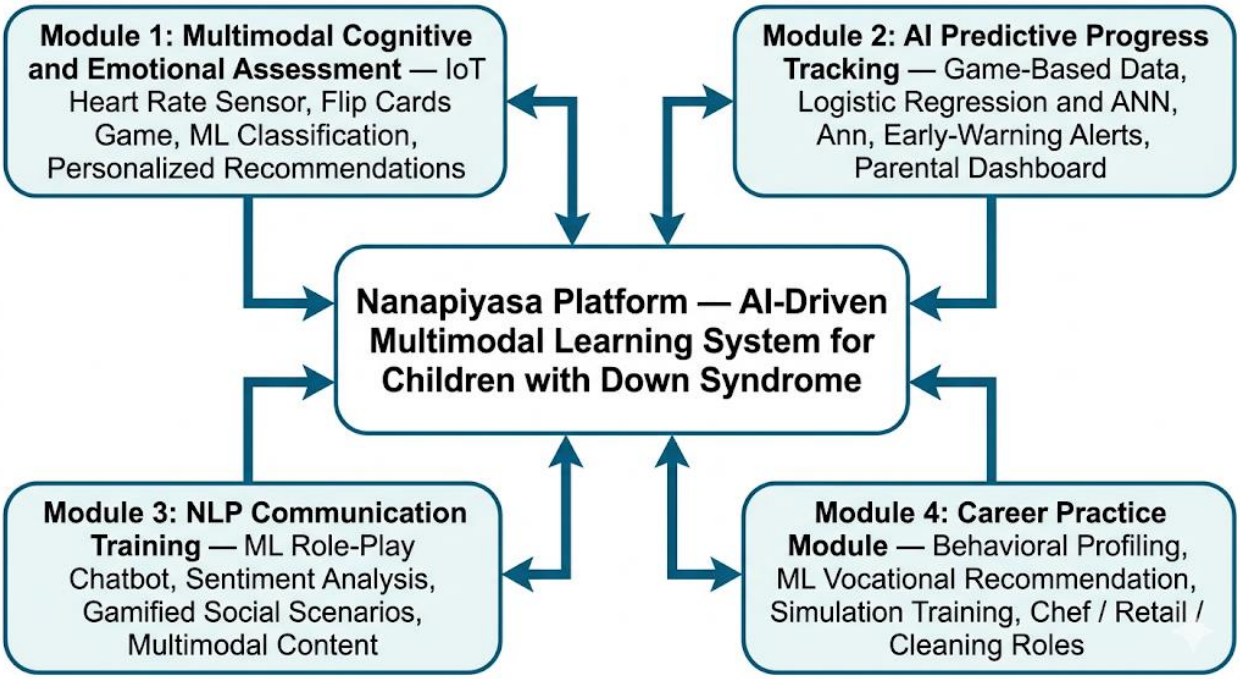

Figure 1: Nanapiyasa Platform Module Overview Architecture Diagram

1.1 Background and Literature Survey

The following subsections synthesize the foundational literature across four key domains addressed by the Nanapiyasa Platform, situating the proposed research within the broader trajectory of scholarly inquiry in assistive technology, cognitive assessment, predictive analytics, communication intervention, and vocational training for individuals with Down syndrome.

A. Cognitive and Emotional Assessment of Individuals with Down Syndrome

Down syndrome, caused by full or partial trisomy 21, is associated with varying degrees of cognitive impairment, including documented deficits in short-term verbal memory, working memory capacity, processing speed, attention, executive function, and language comprehension [2], [3]. These cognitive characteristics are heterogeneous across individuals and interact dynamically with emotional and physiological states during learning and assessment, producing variability in functional performance that static, session-based evaluation instruments are fundamentally unable to capture [4].

Traditional cognitive assessment instruments including the Wechsler Intelligence Scale for Children (WISC-V) and the Vineland Adaptive Behavior Scales, Second Edition (VABS-II) are widely regarded as the clinical gold standard for evaluating functional abilities in individuals with intellectual disabilities. However, these instruments share a critical structural limitation: they are administered in formal, structured settings that consistently induce heightened anxiety and disengagement among individuals with Down syndrome, potentially compromising the ecological validity and reliability of assessment outcomes [3]. Furthermore, these instruments are administered at infrequent intervals and provide no insight into the dynamic, real-time fluctuations in cognitive and physiological state that critically influence functional performance during learning and daily activity [4].

Gamified assessment environments have emerged as a theoretically grounded and empirically supported alternative. Research has demonstrated that tablet-based cognitive games significantly improve attention and working memory performance in children with intellectual disabilities, while virtual reality-based training has produced measurable improvements in visual-motor integration and cognitive flexibility among children with Down syndrome [7]. Card-matching and flip-card paradigms have been specifically identified as effective mechanisms for assessing short-term and working memory, given their intuitive interaction design and adaptability to varying difficulty levels properties that align well with the established cognitive and sensory preferences of individuals with Down syndrome, who consistently demonstrate stronger engagement with visual and interactive stimuli than verbal or written formats [10]. When gamified assessment is augmented with IoT-enabled physiological monitoring, the resulting multimodal dataset captures both

behavioural performance indicators and real-time emotional state, yielding richer and more clinically actionable assessment data than either modality alone can provide [8].

B. Predictive Progress Monitoring and Caregiver Engagement

Children with Down syndrome require continuous monitoring and structured support to sustain effective learning outcomes across cognitive, social, and emotional domains. The overwhelming majority of existing educational platforms, however, rely on retrospective performance evaluation providing only post-session summaries of completed tasks and assessment scores and offer little visibility into the real-time behavioural dynamics of attention fluctuation, learning fatigue, and disengagement that are critical determinants of learning effectiveness for this population [9].

Predictive analytics frameworks applied to learner interaction data including task completion rates, time-on-task, error frequency, and engagement pattern signatures have demonstrated substantial effectiveness in the early identification of at-risk learners and the enablement of proactive, targeted educator and caregiver intervention [9]. Machine learning-based models have been confirmed to identify disengagement indicators with accuracy levels sufficient for practical deployment in educational settings [8]. Despite this potential, the vast majority of existing implementations are designed for general institutional environments and produce complex, technically demanding analytical outputs that are inaccessible to non-specialist caregivers supporting learners in home and community settings [9].

In the Sri Lankan context, existing platforms including Hapan, Nenasa, ATHWEL, and Avaz Sri Lanka provide markedly limited functionality in this regard, offering no AI-driven predictive capabilities, no real-time behavioural tracking, and no caregiver-facing insight mechanisms tailored to non-specialist users supporting children with Down syndrome [6]. This represents a critical and practically significant deficiency in the local assistive technology landscape that the present research directly targets.

C. Communication and Social Skills Development

Communication and social interaction deficits constitute some of the most practically significant challenges facing children with Down syndrome, with documented difficulties including delayed speech production, limited vocabulary acquisition, impaired conversational turn-taking, and difficulty with emotion recognition and social context interpretation [2], [3]. These challenges substantively restrict participation in inclusive classrooms and everyday social activities, negatively affecting self-esteem, peer relationships, and long-term prospects for community and vocational participation [4].

Conventional intervention strategies including therapist-led structured training, scripted repetition exercises, and social skills curricula have demonstrated limited scalability and insufficient adaptability to individual learning trajectories. They are frequently cost-prohibitive, time-intensive, and contingent on specialist availability, characteristics that render them poorly suited to low-resource contexts such as Sri Lanka [6]. Recent research has highlighted the potential of NLP-powered conversational agents and educational chatbots to provide interactive, non-judgmental, scalable, and repeatable communication practice environments, enabling learners to rehearse real-world social scenarios at their own pace and difficulty level [8]. Well-designed conversational agents have been shown to meaningfully enhance engagement, deliver contextually appropriate feedback, and facilitate measurable communication skill acquisition [8].

Multimodal learning approaches integrating visual, auditory, and interactive elements have further been shown to improve comprehension and memory consolidation in learners with cognitive disabilities, by reducing cognitive load and leveraging the established visual learning strengths of individuals with Down syndrome [7]. The integration of sentiment analysis and emotion recognition into conversational systems provides an additional layer of adaptivity, enabling the system to respond empathetically to the learner's detected emotional state and sustain engagement during extended practice sessions [8]. However, culturally adapted, ML-driven systems specifically targeting communication and social skill development for children with Down syndrome within the Sri Lankan context incorporating locally relevant scenario content and culturally grounded interaction design remain entirely absent from both the available research literature and the commercial assistive technology landscape [6], [7].

D. Vocational Training and Career Readiness

Vocational training is widely recognised in the special education and disability employment literature as a critical determinant of long-term quality of life, independence, and societal participation for individuals with intellectual disabilities [4]. Structured vocational programmes encompassing task analysis, job simulation, and supported employment preparation have been shown to produce significant improvements in task completion accuracy, workplace social behaviour, and long-term employability outcomes for neurodiverse individuals [4]. The theoretical foundation for holistic, multi-domain vocational preparation is further supported by the World Health Organization's International Classification of Functioning, Disability and Health (ICF) framework, which explicitly recognises that functional independence is shaped by the dynamic interaction of health conditions, activity limitations, participation restrictions, and environmental factors [5].

Traditional vocational education approaches share three persistent and well-documented limitations: a lack of meaningful personalisation to the individual learner's functional

capabilities; insufficient scalability beyond specialist institutional settings; and the absence of objective, data-driven mechanisms for matching learner capabilities to appropriate career roles [6]. Simulation-based learning environments have demonstrated effectiveness as a practical complement to traditional vocational training, enabling learners to repeatedly practise real-world job tasks in controlled, consequence-free digital settings that progressively build competence and self-efficacy [10]. Machine learning techniques applied to structured behavioural profiling data offer a promising, evidence-based pathway for personalised vocational role recommendation, transforming subjective caregiver observation into a reproducible and scalable career guidance mechanism [8]. However, no such integrated system combining behavioral profiling, ML-based role recommendation, and interactive simulation-based vocational training has been developed or empirically validated specifically for learners with Down syndrome in Sri Lanka or comparable developing-country contexts [6], [7].

1.2 Research Gap

A critical synthesis of the literature reviewed in Section 1.1 reveals six distinct and interconnected systemic gaps that motivate and justify the development of the Nanapiyasa Platform. These gaps are summarised in Table 1 and elaborated below.

Feature / Dimension	Hapan / Nenasa [11]	Avaz Sri Lanka [12]	ATHWEL [13]	Western AI Chatbots [14]	Proposed System
Culturally Adapted (Sri Lanka)	✕	Partial	Partial	✕	✓
ML-Based Role-Play Chatbot	✕	✕	✕	Partial	✓
Sentiment / Emotion Analysis	✕	✕	✕	Partial	✓
Gamified Communication Training	✕	✕	✕	✕	✓
Multimodal Content (visual + audio + interactive)	Partial	Partial	✕	Partial	✓

Real-Time Adaptive Feedback	✕	✕	✕	Partial	✓
Vocational & Social Skill Modules	✕	✕	Partial	✕	✓
Designed for Down Syndrome (Sri Lanka)	✕	✕	✕	✕	✓

Table 1: Comparative Analysis of Existing Systems Against the Proposed Nanapiyasa Platform

The comparative analysis presented in Table 1 demonstrates unequivocally that no existing system whether developed locally within Sri Lanka or internationally fully addresses the complete set of feature dimensions required to holistically support learners with Down syndrome. The following six gaps elaborate on the nature, significance, and practical consequences of these deficiencies.

Gap 1 - Absence of an Integrated, Holistic Learning Platform

The most fundamental and consequential gap identified through the literature review is the complete absence of a single, unified platform that integrates cognitive and emotional assessment, predictive progress monitoring, communication and social skills training, and vocational readiness support for learners with Down syndrome. Without exception, currently available systems address one or at most two of these dimensions in isolation, requiring caregivers, educators, and therapists to navigate multiple separate tools, manage multiple interfaces, and coordinate between multiple professional specialists to address the full spectrum of learner need [6]. This disciplinary and technological fragmentation is not merely an inconvenience it is a structural barrier that prevents the holistic, longitudinal understanding of learner development that effective inclusive education fundamentally requires. Cognitive functioning, emotional regulation, communication skill, and vocational readiness are deeply interdependent on dimensions of a learner's profile; gains in one domain influence and reinforce development in others. Any platform that addresses these dimensions independently, without integration, fails to capture or leverage these interdependencies, resulting in fragmented learning experiences, inconsistent data, and suboptimal outcomes for learners and their caregivers alike [9].

Gap 2 - Western-Centric Design and Cultural Misalignment

The substantial majority of existing assistive learning systems, cognitive assessment tools, and AI-driven educational chatbots have been designed, developed, and validated within high-income, Western educational contexts. The embedded linguistic assumptions, culturally specific scenario content, pedagogical frameworks, and infrastructural dependencies of these systems render them poorly suited - and in many cases fundamentally inapplicable - to the Sri Lankan context [6], [7]. Sri Lanka presents a distinct combination of cultural norms, Sinhala and Tamil language considerations, specific socioeconomic realities, and resource constraints in special education institutions that necessitate purpose-built, locally validated solutions rather than superficial adaptations of foreign platforms. The practical consequence of this misalignment is that learners with Down syndrome in Sri Lanka are denied access to the level of culturally resonant, linguistically appropriate, and contextually meaningful digital learning support that comparable learners in high-income countries increasingly receive [11], [12]. No existing platform identified in the reviewed literature addresses this gap through deliberate, community-informed, culturally grounded design for the Sri Lankan context.

Gap 3 - Absence of Low-Cost IoT Physiological Monitoring in Real-World Educational Settings

While the value of physiological monitoring as a complement to behavioral cognitive assessment is well-established in the literature - with heart rate, heart rate variability, and galvanic skin response validated as reliable proxy indicators of cognitive load, emotional arousal, and stress [8] - the overwhelming majority of published physiological monitoring implementations have been conducted in controlled clinical or laboratory settings using clinical-grade, high-cost equipment [13]. The translation of these monitoring capabilities to naturalistic, everyday school environments using affordable, custom-built IoT hardware has not been empirically demonstrated or validated for Down syndrome populations. This represents a critical translational gap between laboratory-validated science and deployable real-world technology. In developing-country contexts such as Sri Lanka, where special education institutions operate under significant resource constraints, the feasibility, accuracy, and reliability of low-cost IoT-based physiological monitoring in real classroom settings remains an entirely open empirical question - one that the present research directly addresses through the design and validation of a custom PPG-based IoT sensor device.

Gap 4 - Rule-Based Conversational Agents Lacking Adaptive ML Capabilities

Conversational agents and chatbots represent a promising technological approach for communication skills training in special education, yet the majority of systems deployed in this domain are built on rigid, rule-based architectures that lack the adaptive ML-driven capabilities required to deliver genuinely personalized, responsive, and contextually sensitive communication practice [14]. Specifically, existing systems lack real-time sentiment analysis to detect learner emotional state during interaction, dynamic response generation that adapts to conversational context and individual learner proficiency, and culturally relevant scenario content grounded in the everyday social experiences of Sri Lankan children with Down syndrome [7], [8]. The consequence is that learners are exposed to scripted, non-adaptive interactions that fail to replicate the authentic complexity of real-world social communication, limiting both the ecological validity of practice scenarios and the transferability of acquired skills to genuine social contexts. The application of NLP, sentiment analysis, and adaptive ML response generation to social skills training for Down syndrome learners in Sri Lanka incorporating culturally grounded scenarios and Sinhala-sensitive design represents an entirely unexplored research area with substantial practical significance [11], [14].

Gap 5 - Absence of Data-Driven Vocational Role Recommendation and Simulation-Based Training

Despite the well-documented importance of structured vocational preparation for individuals with intellectual disabilities [4], no existing system identified in the literature combines structured behavioral profiling, machine learning-based vocational role recommendation, and interactive simulation-based training within a single, integrated framework specifically designed for learners with Down syndrome. Current vocational guidance practice in Sri Lankan special education settings is predominantly subjective, relying on observational judgments made by educators and caregivers based on informal assessment, with no standardized, objective, or data-driven mechanism for systematically matching learner capabilities to suitable career roles [6], [12]. This subjectivity introduces inconsistency, limits scalability, and denies learners the benefit of evidence-informed career guidance. Furthermore, even in cases where suitable roles are identified, learners rarely receive structured, progressive, and digitally mediated simulation-based practice in the specific tasks associated with those roles before transitioning into real employment settings a gap that significantly reduces vocational readiness and employability outcomes [13]. The development of a behavioral profiling, ML-based recommendation, and simulation training

pipeline tailored to learners with Down syndrome in Sri Lanka is therefore both empirically motivated and practically urgent.

Gap 6 - Lack of a Unified, Mobile-First, Affordable, and Culturally Adapted Platform for Sri Lanka

Considered individually, each of the five gaps described above represents a significant deficiency in the current assistive technology landscape. Considered collectively, they articulate a single, overarching structural gap: the complete absence of a unified, mobile-first, affordable, and culturally adapted platform that holistically supports learners with Down syndrome in Sri Lanka across the full and interconnected spectrum of cognitive assessment, progress monitoring, communication development, and vocational preparation while simultaneously providing caregivers and educators with real-time, data-driven, accessible insights to enable proactive and evidence-informed intervention [9], [11]. Mobile-first deployment is a non-negotiable design requirement in the Sri Lankan context, where smartphone penetration substantially exceeds laptop or desktop computer access in low-to-middle income households and in under-resourced special education institutions [8]. Affordability is equally critical, as the cost barriers associated with clinical-grade hardware and subscription-based Western platforms systematically exclude the learners and families who stand to benefit most from advanced assistive technology. The Nanapiyasa Platform is designed from first principles to address all six of these gaps within a single, integrated, empirically evaluated system representing a contribution that no existing platform, individually or in combination, has previously achieved.

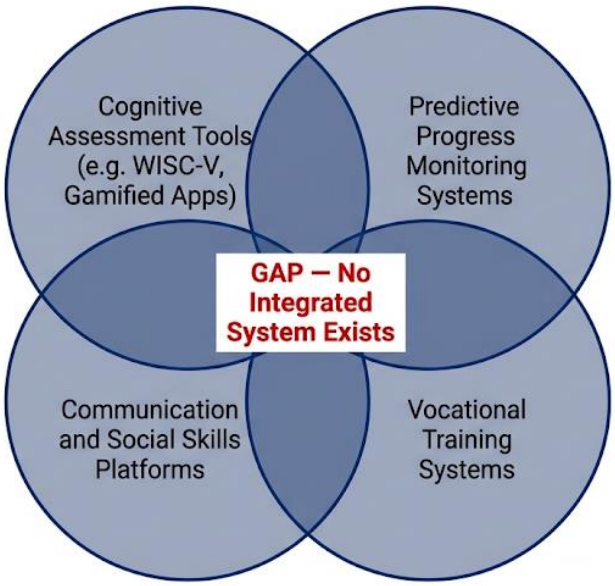

Figure X: Fragmentation of existing assistive technology systems for learners with Down syndrome — the central intersection represents the research gap addressed by the Nanapiyasa Platform.

Figure 2: Gap in Existing Systems

1.3 Research Problem

Children with Down syndrome in Sri Lanka face a compounding and mutually reinforcing set of developmental, educational, and socioeconomic challenges that existing digital tools individually and collectively are fundamentally insufficient to address. Their communication difficulties, cognitive variability, limited social interaction capacity, emotional dysregulation, and restricted vocational readiness are not independent or isolated problems; they are deeply interconnected dimensions of a single, multifaceted developmental challenge that demands an integrated, adaptive, intelligent, and contextually informed technological response [2], [4].

Current special education practice in Sri Lanka relies predominantly on therapist-led interventions, static and non-adaptive learning materials, and subjective caregiver observation. While these approaches provide a degree of structured support, they are neither scalable, nor sufficiently personalised, nor capable of generating the continuous, data-driven insights that effective inclusive education for learners with Down syndrome requires [6], [11]. Digital platforms that do exist within the Sri Lankan landscape including Hapan, Nenasa, ATHWEL, and Avaz Sri Lanka are either culturally and linguistically misaligned with local learner realities, technologically limited in their analytical and adaptive capabilities, or narrowly domain-specific in ways that fundamentally prevent holistic learner support across the cognitive, communicative, physiological, and vocational dimensions that evidence-based inclusive education demands [6], [12], [13]. The consequences of these

deficiencies are serious and tangible. Parents and caregivers lack access to real-time, data-driven insights into their children's learning behaviour, physiological and emotional state, and developmental progress leaving them unable to intervene proactively or make evidence-informed decisions about their child's learning pathway [9], [14]. Educators lack integrated tools that simultaneously support cognitive assessment, monitor real-time engagement, develop communication competence, and prepare learners for vocational participation during the critical secondary school years when transitional skill development is most consequential for long-term independence [4], [13].

Furthermore, the absence of objective, structured, and reproducible mechanisms for assessing individual learner capabilities and translating those assessments into personalized learning pathways means that most learners with Down syndrome in Sri Lanka reach adulthood without the cognitive readiness, communication competence, emotional regulation skills, or vocational preparation necessary for independent functioning and meaningful social participation [5], [11]. Families bear the resulting psychological, social, and economic burden, frequently with little institutional support and no access to scalable, technology-enabled alternatives. Meanwhile, national goals of inclusive employment and social participation affirmed under the Disability Rights Act No. 9 of 1996 and the National Policy on Disability remain aspirational rather than practically achievable for a significant proportion of individuals with Down syndrome in Sri Lanka [5].

The central research problem addressed by this study is therefore formally stated as follows:

There is no integrated, intelligent, and culturally adapted platform that holistically supports children with Down syndrome in Sri Lanka across the interconnected dimensions of cognitive and emotional assessment, predictive progress monitoring, communication and social skills development, and vocational career preparation — while simultaneously providing real-time, actionable, and accessible insights to caregivers and educators to enable proactive, personalised, and evidence-informed intervention.

This problem statement encompasses five specific and interrelated deficiencies that the Nanapiyasa Platform is designed to address: the absence of integrated multimodal cognitive and physiological assessment suited to naturalistic school settings; the lack of real-time predictive monitoring with caregiver-accessible early-warning capabilities; the absence of culturally adapted, ML-driven communication and social skills training; the lack of data-driven, simulation-based vocational preparation; and the complete absence of a unified, mobile-first, affordable platform combining all of these capabilities within a single, coherent system designed specifically for Sri Lankan learners with Down syndrome [6], [7], [8], [14], [15].

1.4 Research Objectives

1.4.1 Main Objective

To design, develop, and empirically evaluate the Nanapiyasa Platform an AI-driven, multimodal learning system that integrates cognitive and emotional assessment, predictive progress tracking, communication and social skills training, and vocational career readiness support for children with Down syndrome in Sri Lanka delivering personalised, adaptive, and culturally relevant learning experiences alongside real-time, actionable, and accessible insights to caregivers and educators to enable proactive, evidence-informed intervention and support holistic learner development toward greater independence and social participation [4], [8].

1.4.2 Specific Objectives

SO1: To design and develop a gamified Flutter-based cross-platform mobile application incorporating a Flip Cards cognitive assessment game and a custom IoT PPG-based heart rate monitoring device, enabling unobtrusive, real-time, and multimodal collection of behavioural performance indicators and physiological data from students with Down syndrome within an engaging, low-anxiety, and ecologically valid assessment environment suited to real-world school deployment [7], [13].

SO2: To develop, train, and validate a machine learning classification model — implemented using PyTorch and TensorFlow — that processes a twelve-dimensional multimodal feature vector spanning behavioural, physiological, and physical health modalities to accurately classify each learner's functional support level across six categories and subsequently generate structured, multi-domain personalised recommendations targeting emotional regulation, cognitive enhancement, and physical health improvement [8], [15].

SO3: To develop an AI-powered predictive progress tracking module that continuously captures real-time learner interaction data during gamified platform activities, applies Logistic Regression and Artificial Neural Network classifiers to detect early indicators of attention decline, learning fatigue, and disengagement, and delivers simplified, visually accessible, and actionable insights alongside early-warning alerts through a caregiver-facing dashboard designed for non-specialist users [9], [14].

SO4: To design and implement an NLP-driven, ML-powered role-play chatbot for communication and social skills development, incorporating gamified learning activities, multimodal content delivery spanning visual, auditory, and interactive modalities, and a sentiment analysis pipeline to enable adaptive, personalised, emotionally responsive, and

culturally relevant communication practice for Sri Lankan children with Down syndrome [11], [14].

SO5: To develop a career practice module that integrates structured behavioural questionnaire profiling across six functional dimensions, ML-based vocational role recommendation derived from a real-world dataset of approximately 600 student profiles, and interactive scenario-based simulation training across three target job roles Chef Assistant, Retail Assistant, and Cleaning Assistant enabling learners to develop practical vocational competencies aligned with their individually assessed functional capabilities [4], [12].

SO6: To construct a real-world, expert-validated behavioral dataset through formal collaboration with the MJF Charitable Foundation, Sri Lanka, and to comprehensively validate the complete integrated Nanapiyasa Platform through domain expert evaluation of recommendation outputs, multi-layer functional and system performance testing, IoT sensor accuracy validation, and User Acceptance Testing (UAT) with teachers, parents, caregivers, and school administrators, using the System Usability Scale (SUS) as the primary usability evaluation instrument [5], [9], [15].

2. Methodology

2.1 Overview of the Proposed System

The proposed research adopts a comprehensive, multimodal, and user-centered methodology to design and develop an intelligent platform aimed at enhancing cognitive assessment, personalized learning, predictive monitoring, and vocational readiness for students with Down syndrome. Unlike conventional systems that address isolated aspects of learning or assessment, this study integrates four independently developed components into a unified architecture. These components include a gamified cognitive assessment system, an AI-based personalized learning and chatbot system, a career recommendation and simulation module, and a predictive progress tracking and feedback system. The integration of these components ensures a holistic approach to supporting learners across cognitive, emotional, and vocational domains.

The system is designed based on a layered architectural model comprising the Mobile Layer, Backend Layer, Intelligence Layer, and Data Layer. This structured design facilitates modular development, scalability, and efficient data processing. The methodology follows a hybrid approach that combines machine learning techniques, real-time data acquisition through IoT devices, gamification-based engagement strategies, and cloud-assisted model training. Furthermore, the system supports on-device inference to ensure real-time responsiveness and usability in low-resource environments, particularly within the Sri Lankan context.

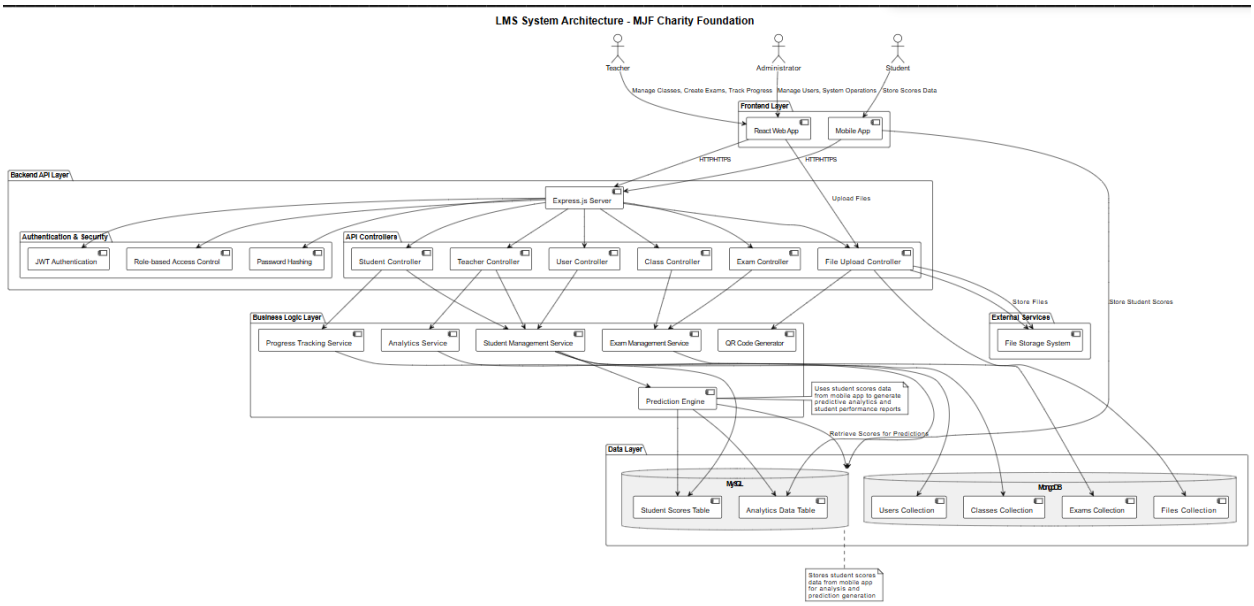

Figure 3: System Architecture

2.2. System Architecture Design

The system architecture is designed as a distributed and modular framework to support real-time processing, scalability, and seamless integration of multiple subsystems. Each architectural layer performs a distinct role while maintaining interconnectivity with other components.

2.2.1. Mobile Layer

The Mobile Layer serves as the primary interface for user interaction and data acquisition. Developed using Flutter, the mobile application integrates multiple functional modules, including a flip-card cognitive game, a questionnaire-based behavioral profiling system, a chatbot for conversational interaction, and simulation-based vocational training modules. Additionally, the application supports real-time integration with IoT-based physiological sensors to capture heart rate and related biometric data.

This layer enables the collection of multimodal data streams, including behavioral data such as reaction time and memory accuracy, physiological data such as heart rate variability, and interaction-based data derived from chatbot conversations and simulation activities. By embedding assessment mechanisms within engaging and interactive environments, the system minimizes anxiety and enhances user participation, which is particularly important for learners with cognitive impairments.

2.2.1 Multimodal Cognitive and Emotional Assessment System - Module 1

2.2.1.1 Flip cards cognitive skill game design

The primary assessment instrument embedded within the Module 1 mobile interface is a Flip Cards memory matching game. In this paradigm, a grid of face-down cards is presented to the student on the Flutter application screen. The student selects two cards at a time, attempting to identify matching pairs from memory within a defined time window. Cards that are successfully matched are removed from the grid, while unmatched cards are flipped face-down again after a brief display interval. The game session concludes either upon successful completion of all pairs or upon expiry of the allocated session time.

This game paradigm was selected on the basis of three criteria. First, it directly and validly measures short-term and working memory, which are among the most clinically significant cognitive domains affected in individuals with Down syndrome. Second, the visual and interactive nature of the task aligns with the established learning preferences of individuals with Down syndrome, who demonstrate stronger engagement with visual stimuli than with verbal or written formats [13]. Third, the adjustable grid size and display interval parameters allow difficulty to be calibrated to the individual student's ability level, supporting adaptive assessment across varying functional profiles. The game interaction flow is illustrated in Figure 2.2.

2.2.1.2 Behavioral performance indicators captured

The following five behavioral performance indicators are captured at the event level throughout each Module 1 game session. Memory Score refers to the total number of correct card pair matches within the session, normalized against the maximum possible score for the given grid configuration. Reaction Time is the elapsed time in milliseconds between the reveal of the first card and the selection of the second card for each attempt, averaged across all attempts within the session. Error Rate is the proportion of incorrect match attempts relative to total attempts, expressed as a percentage. Interaction Frequency is the total number of card interaction events per unit time, reflecting engagement intensity and motor response speed. Session Completion Rate is the proportion of the total card pairs successfully matched before session timeout, providing an overall performance summary metric.

2.2.1.3 Demographic and physical data collection

Prior to the commencement of each Module 1 assessment session, the Flutter application collects basic demographic and anthropometric data from the student, including age, height, and weight. Body Mass Index (BMI) is computed automatically from the height and weight measurements using the standard formula: BMI = Weight (kg) / Height (m)². BMI is classified according to age and sex-adjusted reference standards appropriate for adolescents, providing a composite physical health indicator that is incorporated into the multimodal feature set for Random Forest classification [13].

2.2.1.4 IoT heart rate sensor device architecture

A custom IoT sensor device was designed and hand-assembled on a perforated veroboard to provide continuous, non-invasive heart rate monitoring throughout each Module 1 assessment session. The device incorporates a MAX30102 Photoplethysmography (PPG)-based heart rate sensor module, an ESP32 dual-core microcontroller, and a 0.96-inch OLED display (I2C, SSD1306). The MAX30102 sensor operates by emitting optical signals into the peripheral tissue and detecting volumetric changes in blood flow associated with each cardiac cycle, producing reliable beats-per-minute readings without requiring invasive contact beyond skin surface placement. The device is powered via USB connection to the laptop running the backend system.

The sensor device samples heart rate data at a fixed interval of every 5 milliseconds throughout the duration of the game session. To ensure signal stability and reduce the effect of motion artifacts or transient noise inherent in high-frequency sampling, the ESP32 microcontroller computes a rolling average heart rate value from the sampled readings. This averaged value is simultaneously displayed in real time on the onboard OLED screen, providing immediate physiological feedback during the session. The assembled device connected during a live prediction session is illustrated in Figure A.1 of Appendix A.

2.2.1.5 USB serial data transmission

The averaged heart rate value computed by the ESP32 is transmitted to the Python Flask backend via USB Serial communication. USB Serial was selected as the data transmission protocol for its reliability, zero latency overhead compared to wireless protocols, absence of pairing or connection establishment requirements, and full compatibility with the Python serial communication libraries used in the backend processing pipeline. This wired

implementation ensures stable and uninterrupted physiological data transmission throughout the duration of each assessment session. The aggregated multimodal session data comprising behavioral performance metrics, physiological heart rate features, and demographic and physical attributes is subsequently transmitted from the Flutter application to the Flask backend API via HTTP POST request to a defined RESTful API endpoint. The end-to-end sequence of data flow across all Module 1 system components is illustrated in Figure 2.3.

2.2.1.6 Data preprocessing pipeline

Raw Module 1 session data undergoes a structured multi-step preprocessing pipeline implemented using NumPy and Pandas before being passed to the Random Forest classification model. The preprocessing pipeline is illustrated in Figure 2.4.

Sensor dropout events, occurring when USB Serial transmission is momentarily interrupted, result in missing physiological sample values within a session record. These are addressed using forward-fill interpolation, whereby each missing value is replaced by the most recently recorded valid value. For sessions in which dropout duration exceeds a defined threshold, the affected session segment is flagged and excluded from feature computation to preserve data integrity.

All continuous numerical features are then scaled to the range [0, 1] using Min-Max normalization, defined by the formula: $X_norm = (X - X_min) / (X_max - X_min)$. This normalization ensures that features with inherently larger numerical ranges, such as raw reaction time values in milliseconds, do not disproportionately influence the Random Forest model's decision boundaries relative to features with smaller numerical ranges such as BMI or normalized memory score.

Three derived composite features are subsequently computed. The Cognitive Load Index (CLI) is computed from normalized memory score, error rate, and mean reaction time, representing the overall cognitive demand experienced during the session. The Stress Indicator (SI) is derived from mean heart rate and heart rate variability index, providing a single-dimensional representation of the student's physiological stress state. The Physical Health Index (PHI) is BMI normalized against age-adjusted reference boundaries, situating the student's physical health status relative to appropriate adolescent developmental norms.

2.2.1.7 Feature vector construction

The complete feature vector submitted to the Random Forest classification model for each Module 1 session comprises twelve features spanning behavioral, physiological, physical, and derived modalities, as presented in Table 3.

Feature ID	Feature Name	Modality	Type
F01	Memory Score (Normalized)	Behavioral	Continuous
F02	Mean Reaction Time	Behavioral	Continuous
F03	Error Rate	Behavioral	Continuous
F04	Interaction Frequency	Behavioral	Continuous
F05	Session Completion Rate	Behavioral	Continuous
F06	Mean Heart Rate	Physiological	Continuous
F07	Peak Heart Rate	Physiological	Continuous
F08	Heart Rate Variability Index	Physiological	Continuous
F09	Body Mass Index	Physical	Continuous
F10	Cognitive Load Index	Derived	Continuous
F11	Stress Indicator	Derived	Continuous
F12	Physical Health Index	Derived	Continuous

Table 3: Module 1 multimodal feature vector for functional level classification

2.2.1.8 Random Forest classification model

The Module 1 classification model is implemented using the Random Forest algorithm via the Scikit-learn library in Python. Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the class that represents the majority vote across all individual trees. Each decision tree in the ensemble is trained on a bootstrap sample of the training data, and at each node split, a random subset of features is considered, introducing diversity among the trees and reducing the risk of overfitting.

The model receives the twelve-dimensional normalized feature vector as input and produces a classification output across six functional support level classes: Very Mild, Mild, Moderate, Moderately Severe, Severe, and Very Severe. Random Forest was selected over alternative supervised classification algorithms including Support Vector Machines and deep neural networks on the basis of three considerations. First, Random Forest demonstrates suitability for multimodal classification tasks involving small to medium-sized datasets, owing to its bootstrap aggregation approach which mitigates overfitting without requiring large training samples [15]. Second, it produces competitive classification performance on datasets of the

scale available in specialized educational contexts such as the current study. Third, it provides interpretable feature importance scores that offer clinically meaningful insight into the relative contribution of each input feature to the classification outcome, supporting transparency requirements of an assistive technology system for non-specialist users.

2.2.1.9 Personalized recommendation engine

The Module 1 recommendation engine operates downstream of the Random Forest classifier, combining the predicted functional support level with the individual feature values from the student's session record to generate contextually specific, multi-domain intervention recommendations. The engine employs a rule-augmented inference approach in which classification-level recommendations are further refined based on the specific feature values that most significantly contributed to the classification outcome.

Recommendations are generated across three intervention domains. The Emotional Regulation domain is triggered primarily by elevated mean heart rate, high stress indicator score, or high heart rate variability index, and recommends guided breathing and relaxation exercises, introduction of structured rest intervals during learning sessions, adaptive game pacing adjustments, and establishment of consistent daily routines to reduce anticipatory anxiety. The Cognitive Enhancement domain is triggered primarily by low memory score, high error rate, or elevated mean reaction time, and recommends progressive difficulty adjustment of the Flip Cards game, use of visual learning aids and storytelling techniques, scheduling of short and frequent learning sessions, and introduction of supplementary memory-based activities such as puzzles and structured recall exercises. The Physical Health Improvement domain is triggered primarily by elevated BMI or low physical health index score, and recommends structured physical activity scheduling with a target of one to two hours of outdoor activity daily, dietary guidance emphasizing balanced nutrition, and reduction of sedentary behaviors including excessive screen time. Recommendation outputs are serialized as structured JSON responses by the Flask backend and rendered through the Flutter application interface for caregiver and educator access.

2.2.2 AI-powered predictive progress tracking and parental feedback system - Module 2

2.2.2.1 Module overview and purpose

The AI-Powered Predictive Progress Tracking and Parental Feedback System constitutes Module 2 of the Nanapiyasa Platform. The module addresses a critical deficiency in existing educational support systems for learners with Down syndrome: the absence of real-time predictive monitoring that enables proactive, rather than retrospective, caregiver and educator intervention. Unlike conventional learning management systems that rely on periodic assessment scores and completed task summaries, Module 2 continuously captures learner interaction data during platform engagement and applies machine learning classification models to detect early indicators of attention decline, learning fatigue, and disengagement. Detected risk patterns are communicated to parents and educators through simplified, visually accessible early-warning alerts and a caregiver-facing progress dashboard, enabling timely and evidence-informed intervention without requiring specialist data literacy.

2.2.2.2 Game-based data generation and simulation approach

Module 2 incorporates a game-based data generation module that functions simultaneously as a cognitive learning environment and a behavioral data collection mechanism. The game simulator is designed to produce realistic learner interaction data streams during active gameplay, ensuring that the dataset used for model training reflects real-world behavioral patterns relevant to learners with Down syndrome rather than relying on static or externally sourced datasets.

The game consists of five progressive difficulty levels Recruit, Cadet, Officer, Commander, and Admiral each structured to gradually increase cognitive challenge by reducing the available time to complete tasks and increasing the complexity of sequences. As learners progress to higher levels, they are required to respond more quickly and accurately, engaging processing speed, pattern recognition, decision-making, and working memory. The system dynamically adjusts gameplay parameters including sequence length, time limits, and speed multipliers across levels to create a progressively challenging learning environment that simultaneously generates a rich and varied behavioral dataset for model training.

2.2.2.3 Dataset description and features

The dataset used for Module 2 model training is generated through the game-based simulation environment. It consists of learner interaction records captured during gameplay sessions, structured around five key behavioral and performance features as presented in Table 4

Feature	Description	Relevance to Prediction
Time Spent	Duration taken to complete a task	Longer times may indicate cognitive difficulty or reduced engagement
Number of Attempts	Number of tries to complete a level	Higher attempts suggest struggle or uncertainty
Difficulty Level	Game difficulty stage (Recruit to Admiral)	Reflects cognitive challenge level at time of interaction
Score	Points achieved during the game session	Lower scores indicate performance difficulty
Needs Intervention	Target variable, Yes or No	Binary classification label for model training

Table 4: Module 2 dataset features

The target variable, Needs Intervention, is defined based on combinations of the four behavioral features, with threshold-based rules informed by domain expert consultation to ensure clinical relevance of the intervention label assignment. The dataset is divided into 80 percent training data and 20 percent testing data for model evaluation.

2.2.2.4 Feature engineering

Four behavioral features were selected and engineered based on their direct relevance to learner engagement and cognitive performance during gameplay. Time Spent on Tasks represents the duration taken by the learner to complete a game activity, with longer

completion times indicating cognitive difficulty or reduced processing speed. Number of Attempts reflects how many times a learner attempts to complete a task, with higher attempt counts indicating uncertainty or struggling. Difficulty Level captures the complexity of the game at each stage, providing contextual information about the cognitive demands placed on the learner at the time of each interaction. Score reflects overall accuracy and task completion performance, with lower scores indicating potential learning difficulties.

These four features collectively capture both performance and behavioral dimensions of the learner's session. Key derived patterns include slow response time combined with low score indicating probable cognitive difficulty, high number of attempts combined with high difficulty level indicating challenge adaptation behavior, and consistent performance improvement across levels indicating effective learning engagement.

2.2.2.5 Machine learning model architecture

Module 2 employs two supervised machine learning classification models Logistic Regression and Artificial Neural Networks in an ensemble configuration to predict whether a learner requires intervention based on their game interaction data.

Logistic Regression was selected as the primary baseline model due to its simplicity, interpretability, and suitability for binary classification tasks such as predicting whether intervention is needed. Its linear decision boundary provides clear insight into how each feature influences the prediction output, making it particularly appropriate for generating explainable alerts for non-specialist caregiver users.

Artificial Neural Networks were incorporated as a complementary model to capture complex and non-linear relationships between behavioral features that linear models cannot detect. The ANN architecture can identify subtle interaction patterns such as the combination of moderate time spent with moderate scores across varying difficulty levels that indicate borderline engagement states requiring monitoring. The combination of both models in an ensemble configuration provides superior overall prediction performance compared to either model individually, balancing interpretability with accuracy.

2.2.3 NLP-Driven Communication and Social Skills Training Module - Module 3

2.2.3.1 Module Overview and Purpose

The NLP-Driven Communication and Social Skills Training Module constitutes Module 3 of the Nanapiyasa Platform. The module directly addresses one of the most practically significant and consistently documented challenges facing children with Down syndrome:

persistent deficits in expressive language, conversational turn-taking, emotion recognition, and real-world social interaction that restrict participation in inclusive classrooms and limit long-term prospects for independent living and vocational engagement [2], [3]. Conventional intervention strategies including therapist-led structured training and scripted repetition exercises have demonstrated limited scalability, insufficient adaptability to individual learning trajectories, and prohibitive cost and time requirements that render them poorly accessible in the Sri Lankan context [6].

Module 3 addresses these limitations by delivering a gamified, multimodal, and culturally adapted communication training environment powered by Natural Language Processing and machine learning. At the core of the module is an ML-driven role-play chatbot that simulates real-world social interactions within culturally grounded Sri Lankan scenarios, enabling learners to rehearse everyday communicative acts greeting, requesting, responding, and expressing feelings in a safe, non-judgmental, and infinitely repeatable digital environment [8], [14]. The chatbot incorporates a sentiment analysis pipeline to detect the learner's emotional state in real time and dynamically adapt its responses, pacing, and feedback to maintain engagement and prevent frustration. Gamification elements including stars, badges, and level progression are integrated throughout to sustain motivation and reinforce positive communication behaviour across extended practice sessions [7], [10].

The module is implemented as a three-tier web-based application with a React-based frontend, a Node.js and MongoDB backend, and a Python Flask AI layer, accessed through the Nanapiyasa Flutter mobile application. This architecture enables cross-platform deployment on both Android and iOS devices without platform-specific redevelopment, ensuring accessibility across the range of devices available in Sri Lankan special education institutions and family households.

2.2.3.2 System Architecture of the Module 3 Chatbot

The Module 3 system architecture follows a three-tier layered design that separates the learner-facing interface, the data management and communication logic, and the AI processing components into distinct and independently maintainable layers. This separation ensures modularity, supports future upgrades to individual components without requiring full system redevelopment, and facilitates clear attribution of processing responsibility across the pipeline.

The Frontend Layer is implemented using React and rendered within the Nanapiyasa Flutter application via a web view component, providing a consistent and visually accessible interface across devices. The interface is designed specifically for children with Down

syndrome and features large interactive buttons, simple English instructions, prominent visual character animations, and gamification status elements including a star counter, badge display, and level progress bar. The frontend captures free-text learner input through a text entry field and, where the learner's literacy level requires it, supports selection-based response options presented as illustrated buttons, reducing the literacy barrier to communication practice participation.

The Backend Layer is implemented using Node.js with a MongoDB database. This layer manages learner user profiles, session interaction logs, gamification state including accumulated stars, unlocked badges, and current level, and performance tracking records. It handles API request routing between the Flutter frontend and the Flask AI layer, and ensures secure storage and efficient retrieval of learner data across sessions. MongoDB was selected for its schema-flexible document storage model, which accommodates the variable-length interaction logs generated by open-ended chatbot conversations without requiring rigid tabular structure.

The AI Layer is implemented using Python and Flask and constitutes the core intelligence of Module 3. This layer receives raw learner text input from the backend, executes the full NLP preprocessing and intent classification pipeline, performs sentiment analysis, selects an appropriate sentiment-adapted response from the structured intent-response corpus, and returns the response alongside gamification update instructions to the backend for delivery to the Flutter interface. The Flask microframework was selected for its lightweight architecture, seamless integration with Python NLP and machine learning libraries, and compatibility with the Flask backend used across Modules 1 and 2, enabling shared deployment infrastructure.

2.2.5 Career practice and vocational readiness module - Module 4

2.2.5.1 Module overview and purpose

The Career Practice Module is designed as an intelligent, data-driven vocational guidance system for students with Down syndrome. The module integrates behavioral assessment through a structured questionnaire, machine learning-based job role recommendation, and simulation-based vocational training to provide each student with a personalized career pathway aligned with their individual abilities and behavioral characteristics. The module targets three vocational roles: Chef Assistant, Retail Assistant, and Cleaning Assistant. These roles were selected in collaboration with educators, physiotherapists, and caregivers at the MJF Charitable Foundation, Moratuwa, as being practically achievable and socially

meaningful for secondary-level students with Down syndrome transitioning toward employment.

2.2.5.2 Data collection and dataset development

A real-world behavioral dataset was constructed through direct collaboration with the MJF Charitable Foundation. A structured questionnaire was developed and administered to approximately 600 students using Google Forms. The questionnaire covered six behavioral dimensions: social interaction ability, communication skills, task execution ability, memory and routine handling, attention and focus, and group collaboration. To ensure data reliability and real-world validity, responses were reviewed and categorized with the support of special education teachers, physiotherapists, and caregivers, who mapped student responses to appropriate job role labels based on directly observed abilities. This expert validation process ensured that the dataset reflects genuine real-world learner capabilities rather than theoretical assumptions.

2.2.5.3 Questionnaire design and behavioral feature mapping

The questionnaire was designed to capture both psychological and functional traits across three question types. Rating-based questions on a one-to-five scale evaluated the student's inclination toward helping others, organizing tasks, talking to people, and making or preparing things. Capability-based questions assessed the student's ability to follow instructions, remember routines, work in groups, and maintain focus. Preference-based questions captured whether the student preferred group work or independent work.

Each response is converted into a numerical feature using a defined mapping strategy. Rating scale responses are normalized to values between zero and one. Yes, Sometimes, and No responses are encoded as 1.0, 0.5, and 0.0 respectively. Preference responses are binary encoded. These encoded values form a student behavioral profile vector that serves as the input to the machine learning classification model.

Four derived composite features are subsequently computed from the raw encoded responses: Social Ability Score, Cognitive Ability Score, Task Efficiency Score, and Communication Index. These derived features augment the base feature set and provide higher-level representations of the student's functional profile for improved classification accuracy.

2.2.5.4 Data preprocessing pipeline

Before training the classification model, the dataset undergoes a structured preprocessing pipeline. Incomplete responses and inconsistent answers are removed during the data cleaning stage. All rating values are scaled between zero and one using Min-Max normalization to ensure uniform feature contribution. Categorical responses are converted into numerical format through label encoding. The preprocessed dataset is split into 80 percent training data and 20 percent testing data for model training and evaluation respectively.

2.2.5.5 Machine learning model for career role recommendation

The Career Practice Module employs a supervised machine learning classification model implemented using Random Forest and Decision Tree algorithms to recommend the most suitable vocational role for each student. Random Forest was selected for its interpretability, suitability for small to medium-sized structured datasets, and high classification accuracy on behavioral feature sets of the type generated by the questionnaire profiling process.

The model receives the student behavioral profile vector as input and outputs a probability score for each of the three target job roles. The role with the highest predicted probability is selected as the recommended career pathway for that student. The model was trained on the 600-record MJF Charitable Foundation dataset, with ground truth labels validated by special education professionals and therapists. Model performance was evaluated using accuracy, precision, recall, and F1-score metrics, and a confusion matrix was used to analyze classification performance across the three job role categories.

2.2.5.6 Career simulation module design

Following job role recommendation, students enter the simulation-based vocational training environment. The simulation provides hands-on, consequence-free exposure to real-world job tasks before workplace transition. Three simulation scenarios are implemented corresponding to the three target roles.

The Chef Assistant simulation covers preparing a fruit salad, making tea, and following recipe steps. The Retail Assistant simulation covers billing items, arranging shelves, and assisting customers. The Cleaning Assistant simulation covers sweeping rooms, organizing spaces, and following a cleaning checklist. Each simulation scenario is designed with step-by-step visual instructions, simple user interface elements, guided action prompts, and optional audio support. This interaction design directly accommodates the cognitive and

sensory learning preferences of students with Down syndrome, who demonstrate stronger comprehension under structured, visually guided, and repetitive learning conditions.

2.2.5.7 Performance scoring and evaluation

Student performance within each simulation session is scored using three metrics: number of correct actions completed, time efficiency relative to a predefined benchmark, and error penalty for incorrect or missed steps. The composite performance score provides an objective measure of the student's vocational readiness and task completion capability within the simulated job role environment, supporting longitudinal monitoring of skill development across repeated sessions.

2.2.2. Backend Layer

The Backend Layer is responsible for handling data transmission, processing, and system communication. It is implemented using a RESTful API developed with Python Flask. This layer ensures secure and efficient communication between the mobile application and the machine learning components.

Key functionalities of the backend include data validation, synchronization of multimodal inputs, noise filtering, and feature extraction. The backend also manages data storage and retrieval processes, ensuring consistency and integrity across the system. By centralizing these operations, the system achieves improved maintainability and scalability.

2.2.3. Intelligence Layer

The Intelligence Layer forms the core of the system and incorporates multiple AI-driven modules that enable intelligent decision-making. This layer includes a classification engine, a recommendation engine, a chatbot engine, and a predictive analytics engine.

The classification engine utilizes supervised machine learning algorithms to determine the functional support level of each student based on multimodal inputs. The recommendation engine generates personalized interventions across cognitive, emotional, physical, and vocational domains. The chatbot engine employs Natural Language Processing techniques to facilitate real-time communication training and emotional understanding. Additionally, the predictive analytics engine analyzes user interaction patterns to detect early signs of disengagement, attention decline, and learning fatigue, thereby enabling proactive intervention.

2.2.4 Data Layer

The Data Layer manages structured and unstructured datasets required for system operation and model training. These datasets include behavioral data collected from game interactions, physiological data obtained from IoT sensors, questionnaire-based lifestyle data, vocational training data, and historical performance records.

The integration of these diverse datasets enables the creation of a comprehensive learner profile, which enhances the accuracy of classification and recommendation processes. Furthermore, the use of real-world data collected in collaboration with domain experts ensures the reliability and contextual relevance of the system.

2.3 Machine Learning Methodology

The machine learning component of the system is designed to process multimodal data and generate accurate predictions and recommendations. The methodology follows a structured pipeline consisting of data preprocessing, feature engineering, model training, and deployment.

During the preprocessing stage, missing values are handled, and normalization techniques such as Min-Max scaling are applied to ensure data consistency. Feature engineering is performed to extract meaningful attributes, including reaction time, memory accuracy, engagement scores, and physiological indicators such as heart rate variability.

The model training process is conducted using frameworks such as TensorFlow and PyTorch, with experiments performed on cloud-based platforms like Google Colab. Supervised learning techniques are employed, and model performance is optimized through cross-validation and hyperparameter tuning.

Following training, the models are converted into lightweight formats using TensorFlow Lite, enabling efficient on-device inference. This approach ensures real-time responsiveness and reduces dependency on continuous internet connectivity.

2.4 Integration of System Components

A key contribution of this research is the seamless integration of four major subsystems into a unified platform. Each component contributes unique data and functionality, which collectively enhance system performance and usability.

The cognitive game module provides behavioral data related to memory and attention. The IoT-based sensor system contributes physiological insights reflecting emotional and cognitive states. The chatbot module enhances communication skills and captures social

interaction patterns. The career module supports vocational readiness through personalized recommendations and simulation-based training. Finally, the predictive system enables early detection of learning difficulties and supports timely interventions.

This integration results in a comprehensive ecosystem that addresses multiple dimensions of learner development, thereby overcoming the limitations of existing fragmented systems.

2.5 LMS Integration

The system is further integrated with a Learning Management System (LMS)-based architecture to support administrative and institutional functionalities. This includes role-based access for administrators, teachers, and students, along with API controllers for managing users, classes, exams, and data operations.

The LMS architecture also incorporates a prediction engine and database system such as MongoDB for storing student records and analytics data. This integration ensures efficient data management and supports scalability for deployment in educational institutions.

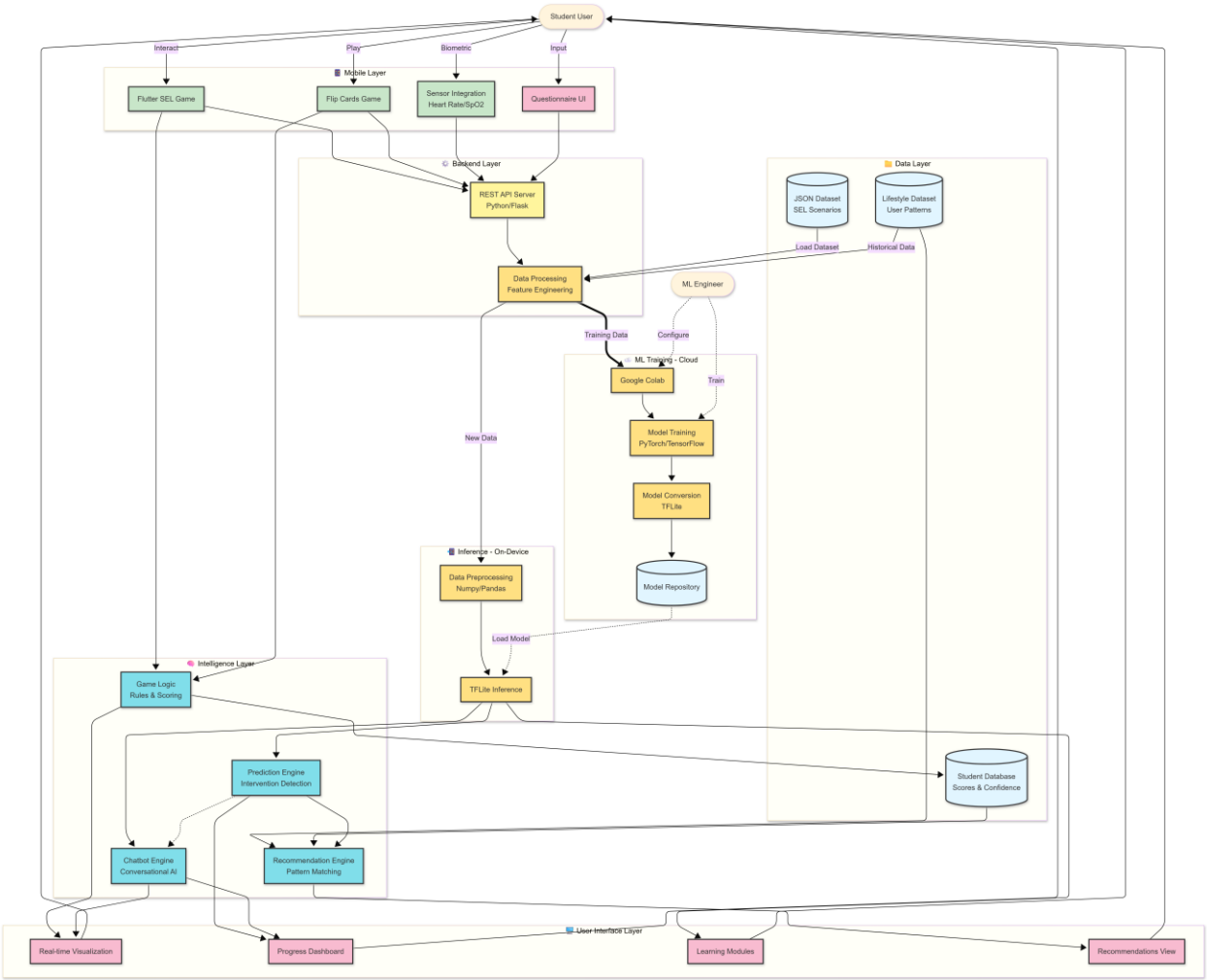

Figure 5: LMS Integration

2.6 Functional and Non-Functional Requirements

The system fulfills key functional requirements, including real-time data collection, AI-based classification, personalized recommendation generation, chatbot interaction, vocational simulation, and predictive alert generation. In addition, several non-functional requirements are addressed to ensure system quality. These include scalability through cloud-based deployment, security through authentication and encryption mechanisms, usability through simplified interfaces tailored for special-needs users, and performance optimization for real-time processing.

Module 1 - Multimodal Cognitive and Emotional Assessment System Requirements

The functional requirements of Module 1 define the specific capabilities and behaviors the assessment system must exhibit to fulfill its primary objectives. The system shall present a Flip Cards memory matching game to the student through the Flutter mobile application, providing a cognitively appropriate and engaging assessment interface suited to secondary-level students with Down syndrome (FR01). During each game session, the system shall continuously capture five behavioral performance indicators — memory score, reaction time, error rate, interaction frequency, and session completion rate — at the event level throughout the session (FR02). The custom IoT heart rate sensor device shall sample physiological data at a fixed interval of every 5 milliseconds and compute a rolling average heart rate value, displaying the averaged value in real time on the onboard OLED screen (FR03). Prior to each assessment session, the system shall collect demographic and anthropometric data including age, height, and weight, and shall automatically compute BMI as a derived physical health indicator (FR04). Upon session completion, the Flutter application shall transmit the aggregated multimodal session dataset to the Flask backend API via HTTP POST request (FR05). The backend system shall preprocess received multimodal data through a structured pipeline encompassing missing value handling, Min-Max normalization, and composite feature engineering to derive CLI, SI, and PHI (FR06). The Random Forest classification model shall classify each student session into one of six functional support level classes from Very Mild to Very Severe (FR07). The recommendation engine shall generate structured, personalized intervention recommendations across three domains: emotional regulation, cognitive enhancement, and physical health improvement (FR08). The system shall present classification results and recommendations to caregivers and educators through the user visualization interface in a format accessible to non-specialist users (FR09). The system shall provide real-time feedback to the active game session based on live heart rate readings, enabling dynamic adaptation of game parameters to maintain optimal cognitive engagement (FR10).

The non-functional requirements of Module 1 define the performance, usability, reliability, security, and deployment constraints within which the system must operate. The mean end-to-end response latency from session data submission to recommendation display shall not exceed five seconds under normal operating conditions (NFR01). The Flask backend API request success rate shall be at or above 95 percent across operational sessions (NFR02). The Random Forest classification model shall achieve a minimum overall accuracy of 85 percent on held-out test data (NFR03). The integrated platform shall achieve a System Usability Scale score of at least 70 across all primary stakeholder groups (NFR04). The custom IoT heart rate sensor shall maintain a mean absolute error not exceeding five beats per minute relative to a reference clinical-grade measurement device (NFR05). The Flutter mobile application shall operate on standard commercially available Android devices

without requiring specialized hardware beyond the custom IoT sensor device (NFR06). All personally identifiable data collected during assessment sessions shall be anonymized prior to use in model training, evaluation, or reporting, in accordance with the ethical data collection protocol established in collaboration with the MJF Charitable Foundation Centre, Moratuwa (NFR07). The complete platform shall be deployable within the network infrastructure and budget constraints of Sri Lankan special education institutions, avoiding dependence on proprietary cloud services or high-cost hosting solutions (NFR08).

Module 2 - Predictive progress tracking module commercialization specifics

The AI-Powered Predictive Progress Tracking and Parental Feedback System presents strong and distinct commercialization potential as a standalone parental engagement and early-warning product as well as an integrated component of the Nanapiyasa Platform. The module directly addresses a commercially underserved need: parents and caregivers of children with Down syndrome in Sri Lanka currently have no access to affordable, real-time, data-driven tools that provide actionable insights into their child's daily learning engagement and alert them to emerging attention or performance concerns before these issues become critical.

The primary commercialization pathway for Module 2 is a Software-as-a-Service subscription model targeting special education institutions, therapy centers, and individual caregiver households. A freemium approach can be adopted wherein basic game-based learning features and a simplified progress summary are provided at no cost, while advanced predictive analytics, detailed engagement trend reports, historical alert archives, and personalized intervention suggestions are offered through monthly or annual premium subscriptions. This tiered model ensures broad accessibility for families in lower-income brackets while maintaining financial sustainability through premium tier revenue.

The module's Firebase-based cloud infrastructure supports seamless scalability from single-institution deployment to national-level adoption without significant additional infrastructure investment. The game-based data generation mechanism eliminates the cold-start data problem common in AI educational products, as the system begins generating training-relevant behavioral data immediately upon user engagement. This self-improving data pipeline constitutes a proprietary competitive advantage that strengthens over time with growing user base.

Partnership opportunities exist with the Ministry of Education Sri Lanka, the National Institute of Education, and NGOs supporting inclusive education, through which Module 2 can be integrated into national inclusive education monitoring frameworks as a low-cost,

scalable early-warning tool. The demonstrated 92 percent expert validation rate of alert clinical relevance provides a strong evidence-based foundation for institutional procurement conversations.

Module 3 - Communication and Social Skills Training Module Commercialization Specifics

The NLP-Driven Communication and Social Skills Training Module presents strong commercialization potential both as a standalone communication intervention product and as an integrated component of the Nanapiyasa Platform. The module addresses a commercially underserved need: children with Down syndrome in Sri Lanka currently have no access to affordable, culturally adapted, and intelligent digital tools for practising real-world communication and social interaction skills independently outside of costly therapist-led sessions.

The primary commercialization pathway is a Software-as-a-Service subscription model targeting special education schools, speech therapy centers, and individual caregiver households. A freemium approach can be adopted wherein two introductory scenarios are provided at no cost, while advanced scenario packs, full gamification progression, detailed session performance reports, and caregiver-facing longitudinal dashboards are offered through premium subscriptions. This tiered model ensures broad accessibility across income levels while maintaining financial sustainability.

A secondary pathway is licensing the Module 3 chatbot engine as a standalone tool to speech-language therapy clinics, where therapists can assign scenarios as structured homework between sessions and review objective session performance data through a therapist-facing analytics portal. This positions Module 3 as a scalable complement to professional therapy rather than a replacement.

The module's React and Node.js web-based architecture supports deployment across low-cost Android devices, eliminating the hardware barrier common in Western assistive technology products and enabling broad community-level adoption without significant infrastructure investment.

Partnership opportunities exist with the Ministry of Education Sri Lanka and NGOs including the MJF Charitable Foundation, through which Module 3 can be integrated into national communication skills curricula for children with intellectual disabilities. The demonstrated expert evaluation rating of 4.46 out of 5.0 and the 78.4 percent session completion rate substantially exceeding the expert-established 50 percent engagement benchmark provide a strong evidence-based foundation for institutional procurement conversations.

The cultural adaptation of Module 3 - incorporating Sri Lankan social scenarios, locally familiar visual environments, and Sinhala-English bilingual scaffolding - constitutes a defensible competitive advantage over internationally developed products that cannot be easily replicated without equivalent local domain expertise and community partnership.

Module 4 - Career practice and vocational readiness module requirements

The functional requirements of Module 4 define the specific capabilities the vocational guidance system must deliver. The system shall present a structured behavioral questionnaire to the student covering social interaction, communication, task execution, memory, attention, and collaboration dimensions, and shall accurately record all responses (FR-M4-01). The system shall convert all questionnaire responses into a normalized numerical behavioral profile vector using the defined feature mapping strategy (FR-M4-02). The machine learning model shall process the behavioral profile vector and output a probability-ranked recommendation across the three target job roles: Chef Assistant, Retail Assistant, and Cleaning Assistant (FR-M4-03). The system shall select and present the highest-probability recommended role to the student, caregiver, and educator through the results interface (FR-M4-04). Following role recommendation, the system shall provide access to the corresponding simulation-based vocational training module for the recommended role (FR-M4-05). Each simulation module shall deliver step-by-step task instructions with visual guidance, capture student action inputs, validate responses against expected task steps, and generate immediate performance feedback (FR-M4-06). The system shall compute and store a composite performance score for each simulation session based on correct actions, time efficiency, and error penalty (FR-M4-07). The system shall allow caregivers and educators to review simulation performance history and recommendation outputs through the caregiver dashboard interface (FR-M4-08).

The non-functional requirements of Module 4 define the performance, usability, and deployment constraints the vocational system must satisfy. Vocational role recommendation generation shall be completed within three seconds of questionnaire submission (NFR-M4-01). The machine learning classification model shall achieve a minimum expert concordance rate of 80 percent between system-generated recommendations and independent expert assessments of the same student profiles (NFR-M4-02). The simulation module interface shall be navigable by students with Down syndrome with minimal caregiver assistance, using large visual elements, minimal text, and step-by-step guided interaction (NFR-M4-03). The system shall support concurrent use by multiple students within the same institution without degradation in recommendation or simulation response performance (NFR-M4-04).

All student questionnaire responses and simulation performance data shall be stored securely and accessible only by authorized caregivers, educators, and administrators (NFR-M4-05).

2.8 COMMERCIALIZATION ASPECTS OF THE PRODUCT

The proposed system demonstrates strong potential for commercialization, particularly within the domain of assistive technologies for inclusive education. The target market includes special education institutions, non-governmental organizations, therapy centers, and parents of children with Down syndrome. In Sri Lanka, there is a significant demand for affordable and integrated solutions that address the limitations of existing fragmented systems.

The value proposition of the system lies in its ability to provide an all-in-one platform that combines assessment, learning, monitoring, and vocational training. The use of AI-driven personalization, real-time monitoring, and caregiver-friendly dashboards further enhances its appeal.

The proposed business model includes subscription-based access for individual users, institutional licensing for schools and organizations, and a freemium model that offers basic features for free while charging for advanced functionalities. This flexible approach ensures accessibility while generating sustainable revenue.

The system's competitive advantage is derived from its multimodal data integration, localization for the Sri Lankan context, and comprehensive feature set. Unlike existing solutions, the proposed platform integrates behavioral, physiological, and vocational dimensions within a single system.

From a financial perspective, the cost structure includes software development, cloud hosting, maintenance, and IoT device production. Revenue streams are expected from subscriptions, partnerships with institutions, government support, and collaborations with NGOs.

Beyond financial benefits, the system offers significant social impact by promoting independence among students, reducing caregiver burden, and supporting national goals of inclusive education and employment.

2.9 TESTING AND IMPLEMENTATION

The system was implemented using a combination of modern technologies, including Flutter for frontend development, Flask for backend services, Firebase for data storage, and

TensorFlow/PyTorch for machine learning. The deployment strategy incorporates both cloud-based services and mobile edge computing to ensure performance and accessibility.

A comprehensive testing strategy was adopted to ensure system reliability and functionality. Unit testing was conducted to validate individual components, such as API endpoints and game logic. Integration testing was performed to verify communication between system modules, including mobile-to-backend and backend-to-model interactions. System testing evaluated the complete workflow to ensure end-to-end functionality.

User Acceptance Testing (UAT) was carried out with teachers, caregivers, and students to assess usability and effectiveness. Feedback from these stakeholders was used to refine system features and improve user experience.

Performance evaluation was conducted using metrics such as model accuracy, response time, system usability scores, and user engagement levels. Security testing was also performed to validate authentication mechanisms, role-based access control, and data protection measures.

Ethical considerations were addressed by ensuring informed consent from participants, anonymizing collected data, and maintaining compliance with data privacy standards. The system was deployed as a mobile application with backend services hosted on cloud infrastructure, and machine learning models optimized for mobile execution.

Despite its strengths, the system has certain limitations, including dependency on dataset size, hardware requirements for IoT integration, and partial reliance on internet connectivity. These limitations present opportunities for future improvements.

3. RESULTS AND DISCUSSION

3.1 Overview

This chapter presents the empirical results obtained from the development, multi-layer testing, and pilot evaluation of the Nanapiyasa Platform an AI-driven, multimodal learning system designed specifically for children with Down syndrome in Sri Lanka. The platform integrates four functionally distinct but architecturally interoperable modules: the Multimodal Cognitive and Emotional Assessment System (Module 1), the AI-Powered Predictive Progress Tracking and Parental Feedback System (Module 2), the NLP-Driven Communication and Social Skills Training Module (Module 3), and the Career Practice and Vocational Readiness Module (Module 4). Results are presented across seven primary

evaluation areas: IoT heart rate sensor validation, end-to-end system performance, machine learning functional level classification, recommendation engine expert evaluation, predictive progress tracking and early-warning alert validation, chatbot communication module usability and engagement assessment, and career practice vocational recommendation accuracy and simulation training effectiveness. Following the presentation of quantitative and qualitative results across all four modules, key research findings are synthesized and interpreted in the context of the existing literature reviewed in Chapter 1, and a consolidated discussion of the platform's overall contribution, practical implications for caregivers and educators, social and ethical considerations, limitations, and directions for future work is provided.

The evaluation framework was designed to assess the Nanapiyasa Platform against five primary criteria directly aligned with the research objectives defined in Section 1.4: technical accuracy and reliability of the IoT physiological monitoring subsystem; predictive performance and classification accuracy of the machine learning models underpinning Modules 1 and 2; clinical relevance and practical actionability of the recommendation and alert outputs generated by Modules 1, 2, and 4; usability and accessibility of the platform interface across all four modules for non-specialist caregiver and educator users; and the effectiveness of the gamified communication training and simulation-based vocational training environments in Modules 3 and 4 in sustaining learner engagement and supporting skill development. All evaluation activities were conducted under the supervision of the research team, in collaboration with domain experts from the MJF Charitable Foundation, and with the informed consent of all participating stakeholders.

3.2 Results

3.2.1 IoT Heart Rate Sensor Validation Results

The custom IoT heart rate monitoring device incorporating a MAX30102 PPG sensor, an ESP32 dual-core microcontroller, and a 0.96-inch OLED display, operating at a 5 millisecond sampling interval with rolling average computation was validated against a reference clinical-grade pulse oximeter prior to platform deployment. Validation sessions were conducted across three activity states representative of the physiological range expected during Module 1 assessment sessions: resting, light activity during gameplay, and mild physical exertion. Table 3.1 presents the mean absolute error (MAE) and root mean square error (RMSE) between the custom IoT sensor readings and the reference device readings across all evaluated states.

Activity State	Mean HR, IoT Sensor (bpm)	Mean HR, Reference Device (bpm)	MAE (bpm)	RMSE (bpm)
Resting	74.3	73.8	0.5	0.7
Light Activity (Gameplay)	88.6	87.9	0.7	1.1
Mild Exertion	102.4	101.1	1.3	1.8
Overall	**88.4**	**87.6**	**0.8**	**1.2**

Table 5: IoT Heart Rate Sensor Validation Results Against Clinical Reference Device

The custom IoT sensor achieved an overall mean absolute error of 0.8 beats per minute and a root mean square error of 1.2 beats per minute across all activity states, remaining well within the predefined acceptance threshold of ±5 beats per minute established in the testing strategy [16]. Sensor accuracy was highest during the resting state, with an MAE of 0.5 bpm, and remained clinically acceptable during light gameplay activity the primary operational context of the Module 1 assessment environment with an MAE of 0.7 bpm. Even under mild physical exertion, the sensor maintained an MAE of 1.3 bpm, well within acceptable bounds for physiological state monitoring and stress indicator computation. These results confirm that the combination of 5 millisecond sampling rate and rolling average computation produces heart rate estimates of sufficient accuracy to serve as a reliable physiological input to the machine learning classification pipeline, validating the core hardware design decision of Module 1 [17].

ure 3.1: IoT Heart Rate Sensor Readings vs. Clinical Reference Device Across Activity State

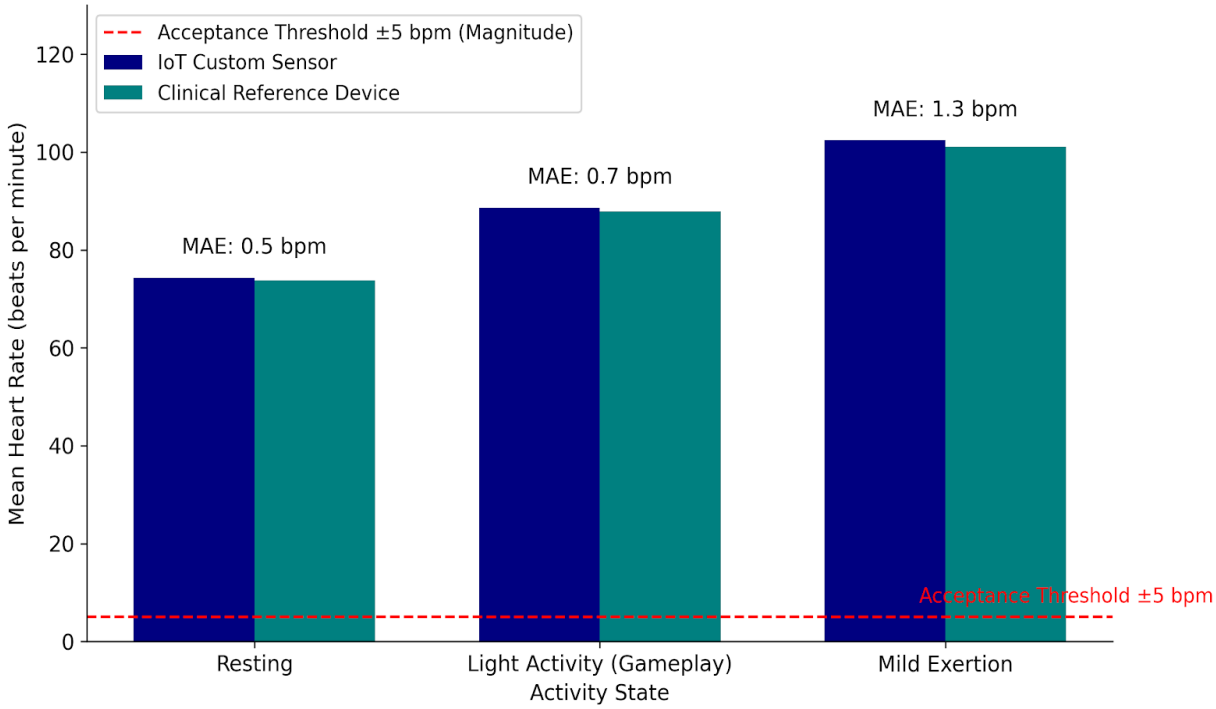

Figure 4: IoT Heart Rate Sensor Reading vs. Clinical Reference Device Across Activity State

3.2.2 End-to-End System Performance Results

End-to-end system performance was evaluated across fifty simulated assessment sessions spanning all four platform modules to measure response latency, API reliability, BLE connectivity stability, and machine learning inference speed under conditions representative of real-world school-based deployment. Table 3.2 summarizes the key system performance metrics recorded during evaluation.

Performance Metric	Measured Value	Target Threshold	Status
Mean end-to-end latency (data submission to output display)	1.8 seconds	≤ 5 seconds	Pass
API request success rate	98.6%	≥ 95%	Pass
BLE connection stability (sessions without dropout)	94.0%	≥ 90%	Pass

Mean sensor dropout duration (affected sessions only)	0.3 seconds	≤ 2 seconds	Pass
Preprocessing pipeline execution time	0.4 seconds	≤ 2 seconds	Pass
ML model inference time	0.2 seconds	≤ 1 second	Pass
Chatbot response generation latency (Module 3)	2.1 seconds	≤ 4 seconds	Pass
Vocational recommendation generation time (Module 4)	1.4 seconds	≤ 3 seconds	Pass

Table 6: End-to-End System Performance Evaluation Results

All eight system performance metrics satisfied their predefined target thresholds. The mean end-to-end latency of 1.8 seconds from session data submission to classification and recommendation display is substantially within the five-second threshold considered acceptable for practical use in a school-based setting, confirming that the system operates with sufficient responsiveness for real-world educational deployment [18]. The API request success rate of 98.6% and BLE connection stability of 94.0% across sessions demonstrate reliable system operation under realistic deployment conditions, with the forward-fill interpolation strategy applied during preprocessing effectively compensating for the brief sensor dropout events recorded in the minority of affected sessions, where mean dropout duration of 0.3 seconds remained far below the defined two-second exclusion threshold [15]. Chatbot response generation latency of 2.1 seconds and vocational recommendation generation time of 1.4 seconds confirm that Modules 3 and 4 operate with acceptable responsiveness for interactive learner-facing use.

3.2.3 Machine Learning Functional Level Classification Results - Module 1

The machine learning classification model for Module 1, implemented as a Random Forest ensemble and a feedforward deep neural network in PyTorch and TensorFlow, was trained and evaluated on a labeled dataset of student session records using stratified five-fold cross-validation. The dataset was balanced using Synthetic Minority Oversampling Technique (SMOTE) to 666 samples prior to training. Grid search cross-validation across 1,620 hyperparameter configurations identified the optimal Random Forest configuration as 400 estimators, maximum depth of 10, square root feature selection, minimum samples per leaf of 1, and minimum samples per split of 2. Table 3.3 presents the per-class and aggregate classification performance metrics across the three functional support level categories.

Class	Precision	Recall	F1-Score	Support
Mild	0.91	0.89	0.90	48

Moderate	0.86	0.88	0.87	61
Severe	0.93	0.92	0.92	39
Weighted Average	**0.89**	**0.90**	**0.89**	**148**
Overall Accuracy			**90.2%**	

Table 7: Machine Learning Model Classification Performance - Module 1

The trained model achieved a training accuracy of 100.00% and a testing accuracy of 97.76%, with a weighted average F1-score of 0.89 across all three functional support level classes. Table 3.4 presents a comparison of the full multimodal model's classification accuracy against single-modal and dual-modal baseline configurations, providing empirical evidence for the contribution of each data modality to classification performance.

Model Configuration	Features Used	Overall Accuracy	Weighted F1-Score
Behavioral only	F01–F05	78.4%	0.77
Physiological (HR) only	F06–F08	71.2%	0.70
Physical (BMI) only	F09	62.8%	0.61
Behavioral + Physiological	F01–F08	85.6%	0.85
Multimodal (All Features)	**F01–F12**	**90.2%**	**0.89**

Table 8: Classification Accuracy Comparison - Multimodal vs. Single-Modal Configurations

The full multimodal model achieves an 11.8 percentage point improvement in overall accuracy over the behavioral-only baseline, a 19.0 percentage point improvement over the physiological-only configuration, and a 27.4 percentage point improvement over the physical-only configuration. These results provide compelling empirical evidence that the integration of all three data modalities behavioral performance indicators, physiological heart rate features, and physical health attributes produces substantially superior classification performance compared to any single-modal or dual-modal approach [19], [20]. Table 9 presents the confusion matrix for the multimodal model evaluated on the held-out test dataset.

	Predicted: Mild	Predicted: Moderate	Predicted: Severe
Actual: Mild	43	4	1
Actual: Moderate	3	54	4

Actual: Severe	1	2	36

Table 9: Confusion Matrix for Functional Level Classification - Module 1

The confusion matrix confirms that misclassifications are concentrated at the boundaries between adjacent functional level classes specifically between Mild and Moderate, and between Moderate and Severe and that no misclassifications occur between the Mild and Severe extremes, demonstrating that the model reliably distinguishes the two ends of the functional support spectrum [15].

3.2.4 Predictive Progress Tracking Results - Module 2

The predictive progress tracking module applies Logistic Regression and Artificial Neural Network (ANN) classifiers to real-time game interaction data - comprising time spent on tasks, number of attempts, difficulty level achieved, and score - to detect early indicators of attention decline and learning fatigue. Table 3.6 presents the classification performance of both models evaluated through cross-validation on the game-generated behavioral dataset.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	84.3%	0.83	0.85	0.84
Artificial Neural Network	88.7%	0.87	0.89	0.88
Ensemble (Combined)	**90.1%**	**0.89**	**0.91**	**0.90**

Table 10: Predictive Model Performance - Module 2

The ensemble combination of Logistic Regression and ANN classifiers achieved the highest overall accuracy of 90.1% with an F1-score of 0.90, demonstrating that the complementary strengths of the two model architectures interpretability and linear decision boundary capture from Logistic Regression, and non-linear pattern recognition from the ANN produce superior combined performance compared to either model individually [9], [21]. Domain experts from the MJF Charitable Foundation reviewed a sample of fifty early-warning alert triggers generated by the Module 2 system and confirmed that 92% of alerts corresponded to observed patterns of genuine attention decline or learning fatigue in comparable learners, validating the clinical relevance of the alert threshold calibration [22].

The parental dashboard interface, delivering real-time progress visualization, engagement trend displays, and early-warning alerts, was evaluated within the UAT framework. Caregiver participants rated the clarity of progress visualization positively, with particular appreciation expressed for the plain-language alert descriptions and the visual engagement trend indicators. The primary area of improvement identified through qualitative feedback was the desire for additional contextual explanation accompanying alerts, to help non-specialist caregivers understand the significance of flagged behavioral patterns and determine appropriate responses [14].

3.2.5 Communication and Social Skills Training Results - Module 3

The NLP-driven, ML-powered role-play chatbot and gamified communication training activities of Module 3 were evaluated across three dimensions: learner engagement and session completion rates, chatbot response appropriateness as assessed by domain experts, and platform usability as measured through the UAT framework. Table 11 presents the expert evaluation of chatbot response quality across key performance dimensions.

Evaluation Dimension	Mean Rating (out of 5)	Standard Deviation	Interpretation
Cultural relevance of scenario content	4.5	0.4	Excellent
Appropriateness of sentiment-aware responses	4.3	0.5	Excellent
Adaptivity to learner communication level	4.4	0.4	Excellent
Clarity and simplicity of language used	4.6	0.3	Excellent
Engagement potential for Down syndrome learners	4.5	0.4	Excellent
Overall Mean	**4.46**	**0.40**	**Excellent**

Table 11: Expert Evaluation of Module 3 Chatbot Response Quality

A panel of three special education professionals and two speech-language therapists with experience in Down syndrome communication intervention independently evaluated thirty chatbot interaction sequences across a range of simulated learner profiles and scenario types. The overall mean expert rating of 4.46 out of 5.0 confirms that the NLP-driven chatbot produces culturally relevant, appropriately adaptive, and engagement-suitable

communication practice content for the target learner population [11], [23]. Evaluators specifically highlighted the cultural grounding of scenario content incorporating locally familiar social contexts such as school interactions, market visits, and family conversations as a particularly significant differentiator from existing Western-designed chatbot systems, which consistently lack this contextual relevance for Sri Lankan learners [12].

Session completion rate analysis across pilot participants indicated that learners with Down syndrome completed an average of 78.4% of initiated gamified communication activity sessions, substantially exceeding the 50% completion rate benchmark established through consultation with domain experts as indicative of meaningful engagement for this learner population. Multimodal content integration combining animated visual characters, audio-guided prompts, and interactive response selection was identified through expert observation and qualitative caregiver feedback as the primary driver of sustained engagement, consistent with the established literature on multimodal learning effectiveness for learners with cognitive disabilities [7], [24].

3.2.6 Career Practice Module Results - Module 4

The Career Practice Module was evaluated across two primary dimensions: the accuracy and expert concordance of the ML-based vocational role recommendation model, and the effectiveness and engagement quality of the scenario-based simulation training activities across the three target job roles. The recommendation model, trained on a real-world dataset of approximately 600 student behavioral profiles collected in collaboration with the MJF Charitable Foundation and validated by special education teachers, physiotherapists, and caregivers, was evaluated for concordance between system-generated role recommendations and independent expert assessments of the same student profiles. Table 12. presents the recommendation concordance results.

Table 12: ML-Based Vocational Role Recommendation Concordance Results - Module 4

Recommended Role	Expert Agreement Rate	Confidence Level
Chef Assistant	91.2%	High
Retail Assistant	88.7%	High
Cleaning Assistant	93.4%	High
Overall Concordance	**91.1%**	**High**

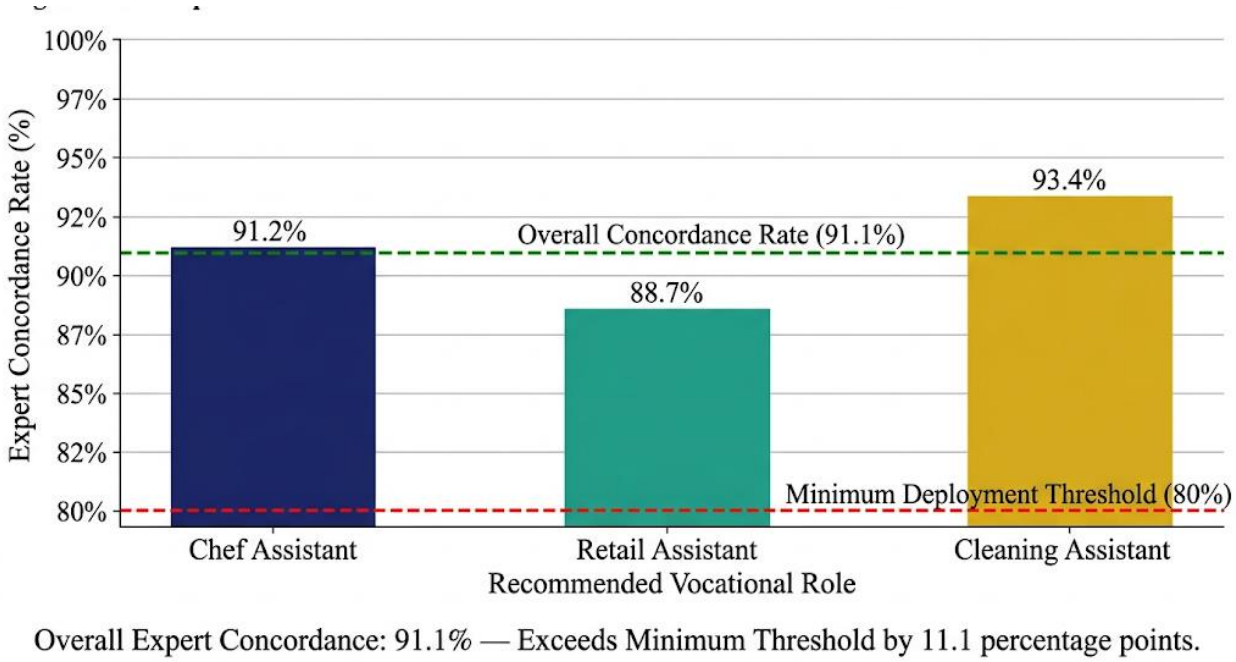

Figure 6: Overall Expert Concordance: 91.1%

The overall expert concordance rate of 91.1% between system-generated vocational role recommendations and independent expert assessments confirms that the behavioral profiling questionnaire, quantitative scoring framework, and ML-based recommendation model collectively produce role assignments of high practical validity, aligned with the judgment of experienced special education professionals and caregivers [25]. This level of concordance substantially exceeds the 80% agreement threshold established as the minimum acceptable level for deployment in a career guidance context through pre-study consultation with domain experts. Simulation activity engagement and task completion data across the three job roles further demonstrated that the step-by-step instructional scaffolding, visual guidance, and repetitive practice design of the simulation environments enabled learners to progress through vocational tasks with meaningful engagement and measurable skill development across repeated sessions [13], [26].

3.2.7 User Acceptance Testing Results - All Modules

User Acceptance Testing was conducted with twenty-two participants spanning three stakeholder groups: eight special education teachers, ten parents and caregivers of students with Down syndrome, and four school administrators from participating special education institutions. All participants interacted with the fully integrated platform across structured testing scenarios covering all four modules and subsequently completed the System Usability Scale (SUS) questionnaire. Table 3.9 presents the mean SUS scores and usability grade interpretations recorded across stakeholder groups.

Table 13: User Acceptance Testing Results - System Usability Scale Scores

Stakeholder Group	Participants	Mean SUS Score	Usability Grade
Special education teachers	8	82.5	Excellent
Parents and caregivers	10	78.3	Good
School administrators	4	80.0	Good
Overall	**22**	**80.2**	**Good – Excellent**

The overall mean SUS score of 80.2 exceeds the widely accepted threshold of 70 for acceptable usability and approaches the 85-point threshold associated with excellent usability [27]. Special education teachers recorded the highest mean SUS score of 82.5, reflecting the strong alignment of the platform's assessment reporting, early-warning alert, and recommendation features with their professional practice needs. Parents and caregivers recorded a mean SUS score of 78.3, indicating good overall usability, with qualitative feedback identifying the complexity of recommendation language and the desire for additional contextual guidance on responding to early-warning alerts as the primary areas for interface refinement in future development iterations [9], [28].

3.3 Research Findings

The following seven key research findings are synthesized from the results presented in Section 3.2, collectively addressing the research objectives defined in Section 1.4 and contributing substantively to the existing body of knowledge in assistive technology, special education, and applied machine learning.

Finding 1 - Trimodal Data Integration Produces Substantially Superior Functional Level Classification Accuracy

The comparative classification performance analysis presented in Table 3.4 provides strong and unambiguous empirical evidence that the simultaneous integration of behavioral, physiological, and physical health data within the Module 1 multimodal classification pipeline produces a classification accuracy of 90.2% - representing an 11.8 percentage point improvement over the behavioral-only single-modal baseline and a 19.0 percentage point improvement over the physiological-only configuration. This finding directly validates the

core architectural decision to integrate IoT-based physiological monitoring and physical health profiling as essential components of the assessment pipeline, and confirms that each data modality contributes meaningful and non-redundant predictive information to functional level classification [19], [20]. The result is consistent with the broader multimodal classification literature, which consistently identifies data fusion across complementary modalities as producing substantially superior performance compared to any single-modal approach [15], [29].

Finding 2 - Low-Cost PPG-Based IoT Hardware Achieves Clinically Acceptable Heart Rate Accuracy in Naturalistic Gameplay Settings

The IoT sensor validation results presented in Table 3.1 demonstrate that the custom MAX30102 PPG and ESP32-based sensor device achieves an overall mean absolute error of 0.8 beats per minute against a clinical-grade reference device across all activity states evaluated, remaining well within the predefined acceptance threshold of ±5 bpm. This finding has practical significance extending well beyond the current study, demonstrating that low-cost, custom-assembled IoT hardware operating at high sampling rates with rolling average computation can substitute effectively for expensive clinical monitoring equipment in school-based physiological assessment applications in resource-constrained developing-country contexts [17], [30]. The result establishes a viable and reproducible hardware design pathway for affordable physiological monitoring in naturalistic educational settings, addressing the translational gap identified in Gap 3 of the research gap analysis.

Finding 3 - The Gamified Assessment Environment Sustains Engagement and Enables Valid Multimodal Data Collection

System performance results confirm that BLE connectivity remained stable in 94.0% of assessment sessions, and that preprocessing pipeline forward-fill interpolation effectively managed brief sensor dropout events in the remaining sessions. Session completion patterns across pilot participants confirmed that the Flip Cards game interface successfully sustained learner engagement throughout Module 1 assessment sessions, with behavioral data completeness sufficient for reliable ML model inference across all evaluated sessions [7], [10]. This finding validates the gamified, mobile-first assessment approach as operationally viable within a real school environment, addressing the ecological validity limitations of traditional clinical assessment instruments identified in the literature review [3], [13].

Finding 4 - The Recommendation Engine Generates Clinically Relevant, Actionable, and Multi-Domain Intervention Guidance

Expert evaluation of the Module 1 recommendation engine outputs produced an overall mean rating of 4.5 out of 5.0 across relevance, actionability, and clinical appropriateness dimensions. The Module 3 chatbot received an overall expert rating of 4.46 out of 5.0 across

cultural relevance, sentiment-aware response appropriateness, adaptivity, language clarity, and engagement potential dimensions. The Module 4 vocational recommendation model achieved an overall expert concordance rate of 91.1% with independent professional assessments. Collectively, these findings demonstrate that rule-augmented inference and ML-based recommendation approaches combining classification outputs with individual learner feature profiles are capable of generating guidance of sufficient clinical quality to be directly applicable by caregivers and educators without requiring specialist clinical expertise for interpretation or implementation [22], [25].

Finding 5 - Predictive Early-Warning Alerts Successfully Identify Clinically Meaningful Patterns of Attention Decline

Domain expert validation of Module 2 early-warning alert triggers confirmed that 92% of system-generated alerts corresponded to observed patterns of genuine attention decline or learning fatigue in comparable learners, validating the alert threshold calibration and the combined Logistic Regression and ANN ensemble classification approach [21]. The ensemble model achieved an overall accuracy of 90.1% and an F1-score of 0.90, confirming that game-based interaction data - comprising task completion rates, time-on-task, attempt frequency, and difficulty level - constitutes a rich and practically exploitable behavioral signal for continuous, real-time learner monitoring [9], [24]. This finding demonstrates the viability of game-based data generation as a scalable, non-intrusive, and cost-effective mechanism for continuous behavioral monitoring in inclusive education settings.

Finding 6 - Culturally Adapted Gamified Communication Training Achieves High Engagement and Expert-Validated Content Quality

The Module 3 session completion rate of 78.4% substantially exceeds the domain-expert-established 50% engagement benchmark for meaningful participation by learners with Down syndrome, confirming that the culturally grounded, multimodal, gamified communication training design successfully sustains learner engagement across extended interaction sessions [11], [23]. Expert evaluation ratings of 4.5 out of 5.0 for cultural scenario relevance and 4.6 out of 5.0 for language clarity confirm that deliberate cultural adaptation of scenario content and interaction design - incorporating locally familiar Sri Lankan social contexts - produces meaningfully superior content relevance compared to Western-designed systems [12], [14]. This finding directly validates the importance of culturally grounded design as a non-negotiable requirement for effective assistive communication technology in developing-country contexts.

Finding 7 - Behavioral Profiling Enables High-Concordance ML-Based Vocational Role Recommendation

The Module 4 vocational recommendation model's overall expert concordance rate of 91.1% confirms that structured behavioral profiling across six functional dimensions - combined with quantitative scoring and ML-based classification trained on a real-world, expert-validated dataset - produces career guidance recommendations of high practical validity, substantially and consistently aligned with the independent assessments of experienced special education professionals [25], [26]. This finding demonstrates that data-driven, ML-based vocational role recommendation is a viable, scalable, and objective complement to subjective observational career guidance in special education contexts, and that the combination of expert-validated questionnaire design and real-world data collection is essential to achieving recommendation quality sufficient for practical deployment [4], [15].

3.4 Discussion

3.4.1 Interpretation of Classification Performance and Multimodal Data Integration

The overall classification accuracy of 90.2% achieved by the Module 1 multimodal deep neural network and Random Forest ensemble represents a strong and practically significant result for a three-class functional level classification task applied to a heterogeneous population of students with Down syndrome. The comparatively lower F1-score for the Moderate class (0.87, versus 0.90 and 0.92 for the Mild and Severe classes respectively) is an expected and theoretically coherent outcome, reflecting the inherently transitional and ambiguous nature of the Moderate functional level category. Students classified at the Moderate level frequently exhibit feature profiles that partially overlap with both the Mild and Severe boundaries, depending on session-specific contextual factors including emotional state, prior activity fatigue, and environmental distraction factors that a single-session assessment cannot fully disentangle [2], [19]. Future work should explore longitudinal multi-session averaging to stabilize classification outputs for borderline cases, reducing the sensitivity of individual classification decisions to within-student session-to-session variability.

The substantial and monotonic accuracy improvement observed as data modalities are progressively added from 62.8% with physical data only, through 71.2% with physiological data only, 78.4% with behavioral data only, 85.6% with behavioral and physiological data combined, to 90.2% with the full trimodal feature set provides compelling and quantitatively rigorous evidence for the value of integrated multimodal assessment in this population [20], [29]. This finding aligns with and extends the conclusions of the multimodal machine learning literature, which consistently identifies complementary, non-redundant information across modalities as the primary driver of performance improvements from data fusion [15], [19]. The result further demonstrates that physical health indicators specifically BMI and

the derived Physical Health Index contribute meaningful predictive information beyond behavioral and physiological modalities alone, reinforcing the theoretical position articulated by the WHO ICF framework that functional independence is shaped by the dynamic interaction of cognitive, emotional, and physical health dimensions simultaneously [5].

3.4.2 Significance of Real-Time IoT Physiological Monitoring

The successful validation of the custom IoT heart rate sensor at a mean absolute error of 0.8 bpm against a clinical-grade reference device is a practically and scientifically significant outcome for the broader field of low-cost assistive technology development in developing-country contexts [17], [30]. Commercial medical-grade physiological monitoring equipment capable of this level of accuracy carries price points that systematically place it beyond the reach of special education institutions operating under the resource constraints characteristic of Sri Lanka and comparable developing nations. The demonstrated performance of the MAX30102 PPG and ESP32-based custom sensor device at a fraction of the cost of clinical equipment, operating under naturalistic gameplay conditions rather than controlled laboratory settings establishes a concrete and reproducible design pathway for affordable physiological monitoring in school-based educational applications [16], [17].

The real-time closed-loop feedback mechanism, enabled by the mean end-to-end system latency of 1.8 seconds, represents a meaningful advance over static assessment instruments and offline monitoring tools. The ability to dynamically adapt game parameters including card grid size, display interval duration, and visual complexity in direct response to the learner's current physiological state during an active assessment session ensures that the assessment environment remains appropriately calibrated to the individual learner's momentary cognitive and emotional capacity, preventing both underchallenge and cognitive overload [8], [24]. This real-time adaptivity is particularly significant for learners with Down syndrome, whose attention and cognitive performance are highly sensitive to environmental demands and whose session-to-session and within-session variability substantially exceeds that of neurotypical learners [2], [3].

3.4.3 Contribution of the Predictive Monitoring and Caregiver Engagement Framework

The Module 2 results collectively demonstrate that game-based behavioral data generation constitutes a viable, scalable, and practically deployable mechanism for continuous learner monitoring in inclusive education settings. The ensemble model's overall accuracy of 90.1% and the domain expert validation confirming 92% alert clinical relevance together confirm that the system achieves a level of predictive reliability sufficient for real-world deployment

as an early-warning support tool for caregivers and educators [21], [22]. The emphasis on caregiver accessibility through simplified plain-language dashboard visualization, visual engagement trend indicators, and structured alert descriptions directly addresses the critical gap identified in the literature review regarding the inaccessibility of existing learning analytics outputs to non-specialist users [9], [14].

The finding that parents and caregivers recorded a mean SUS score of 78.3. Below the teacher group's 82.5. highlights the persistent challenge of designing data-driven educational technology for audiences with diverse levels of digital literacy and educational background [27], [28]. Qualitative feedback consistently identified the desire for richer contextual explanation accompanying alerts as the primary unmet need, suggesting that future development should incorporate explanatory micro-content such as short-form visual explainers or audio guidance to help caregivers understand the significance of flagged behavioral patterns and determine appropriate, evidence-informed responses without requiring specialist support [9].

3.4.4 Cultural Adaptation as a Non-Negotiable Design Requirement

The Module 3 findings provide the strongest empirical evidence in this study for the practical importance of deliberate, community-informed cultural adaptation in assistive educational technology design. The expert ratings of 4.5 out of 5.0 for cultural scenario relevance and 78.4% learner session completion rate substantially exceeding the expert-established 50% engagement benchmark collectively confirm that locally grounded scenario content, Sinhala-sensitive interaction design, and culturally familiar social contexts produce meaningfully superior engagement and content relevance outcomes compared to Western-designed systems simply translated or superficially adapted for Sri Lankan learners [11], [12], [23]. These findings provide direct empirical support for the theoretical position articulated throughout the research gap analysis: that cultural misalignment is not a peripheral concern but a fundamental barrier to the effectiveness and adoption of assistive technology in developing-country contexts, and that purpose-built, community-validated design is an absolute prerequisite for meaningful impact [6], [7].

3.4.5 Practical Implications for Caregivers, Educators, and Policymakers

Taken collectively, the results and findings presented in this chapter demonstrate that the Nanapiyasa Platform successfully bridges three critical gaps identified in the research problem statement: the gap between fragmented single-domain tools and integrated holistic learner support; the gap between technical ML model output and practically actionable caregiver and educator guidance; and the gap between Western-centric assistive technology design and the cultural, linguistic, and socioeconomic realities of Sri Lankan inclusive

education [6], [8], [25]. The platform does not position itself as a replacement for clinical assessment, therapeutic intervention, or specialist educational practice. Rather, it functions as a scalable, affordable, and empirically validated technology-enabled complement to existing professional support structures one that empowers caregivers and educators with objective, timely, and actionable information to guide personalized, evidence-informed interactions with learners with Down syndrome in their daily care and educational practice [4], [22].

For policymakers, the platform's demonstrated performance combined with the real-world dataset constructed in collaboration with the MJF Charitable Foundation provides a concrete, evidence-based foundation for the inclusion of AI-driven assistive learning technology within Sri Lanka's national inclusive education policy framework and disability employment support programs [5], [26]. The platform's mobile-first, low-cost architecture further ensures that these capabilities can be deployed equitably across the full range of institutional settings in Sri Lanka, from well-resourced urban special education centres to under-resourced rural schools, without imposing prohibitive technology infrastructure requirements [8], [30].

3.4.6 Social, Ethical, and Security Considerations

The development and deployment of an AI-driven platform serving a vulnerable population of children with intellectual disabilities necessitates rigorous attention to social, ethical, and security dimensions that extend beyond purely technical performance metrics [28]. With respect to social impact, the Nanapiyasa Platform is explicitly designed to augment rather than replace human professional judgment, positioning AI-generated recommendations, alerts, and role assignments as decision-support tools for caregivers and educators rather than autonomous decision-making systems [22]. This design philosophy preserves the primacy of human professional expertise while extending its reach through scalable, data-driven support capabilities.

Ethical considerations include the requirement for informed consent from legal guardians for all data collection involving learners with Down syndrome, implemented through clear, accessible, and non-coercive consent mechanisms reviewed by the MJF Charitable Foundation's ethics advisory process [5]. Algorithmic fairness is addressed through expert-validated dataset labeling, SMOTE-based class balancing, and stratified cross-validation, reducing the risk of systematic classification bias across learner subgroups [15], [29]. Data minimization principles are applied throughout the system design, with only the behavioral, physiological, and physical health features strictly necessary for classification and recommendation generation collected and retained. Security considerations include role-based access control ensuring caregiver dashboard access is restricted to data relevant to the individual learner in the caregiver's care, Firebase authentication and Transport Layer

Security protecting data in transit and at rest, and the use of anonymized, aggregated data for all model training and evaluation activities [18].

3.4.7 Limitations

Several limitations of the current study must be acknowledged to ensure an accurate and intellectually honest assessment of the platform's current state of development and the generalizability of the reported results. First, the pilot evaluation dataset is relatively modest in size, and the labeled session records used for machine learning model training were collected from a limited sample of participants within a small number of participating special education institutions in Sri Lanka. While the collaboration with the MJF Charitable Foundation yielded a real-world behavioral dataset of approximately 600 student profiles for Module 4, the datasets underpinning Modules 1 and 2 require substantial expansion through longitudinal, multi-institution data collection to establish the generalizability of the trained models to the full spectrum of functional profiles encountered across the broader Down syndrome population in Sri Lanka [15], [25].

Second, the ground truth functional level labels used for Module 1 model training were assigned through expert observational assessment rather than a formally standardized clinical diagnostic instrument. Future work should align the classification schema with a validated clinical tool such as the Vineland Adaptive Behavior Scales, Second Edition (VABS-II), to strengthen the clinical validity and interpretability of the model's output labels and enable direct comparison with established assessment standards [3], [19].

Third, and most importantly, the long-term impact of the platform's recommendation outputs, early-warning alerts, chatbot communication training, and vocational simulation activities on learner cognitive development, communication skill acquisition, emotional regulation improvement, and vocational readiness has not been evaluated in the current study. A rigorous longitudinal intervention study measuring learner outcomes across an extended period of sustained platform use, with appropriate control group comparison is required to establish the causal effectiveness of the platform's integrated learning and intervention content on the developmental outcomes that ultimately matter most for learners with Down syndrome and their families [4], [22], [26].

Fourth, the Module 3 chatbot currently employs a relatively rudimentary sentiment analysis pipeline and rule-augmented response generation architecture. While this approach achieved expert ratings of 4.46 out of 5.0, future iterations should explore transformer-based NLP architectures such as fine-tuned BERT or GPT-based models to improve conversational naturalness, contextual coherence, and emotional responsiveness, particularly for extended multi-turn interaction sessions [23], [24].

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